数理分析方法

连续对应与 Berge 最大值定理

课件编号：108-ZSM-PPT　原始 PPT：113 页

逐页保留原始 PPT 画面 + 中文零基础自学讲解 + 证明专题

阅读说明：本讲义每一页原课件均以原始 PDF 页面完整嵌入；其后的中文解读旨在帮助初学者识别定义、假设、论证步骤。章末为从零展开的核心证明与复习提示。讲义属于学习辅助材料，不代替原教师课件。参考：Gérard Debreu (1959), *Theory of Value*。

目录

1 学前导读

这一讲学什么

对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

读证明先辨认值集是否非空、紧、闭，以及定义域和目标空间的拓扑；缺任何一个假设都可能令结论不成立。

证明题通用方法（先读本页）

严格证明的四个必要动作

第一：写出假设。比如“K 非空、闭、有界”，不能只写“K 很好”。

第二：逐字展开定义。写出 $\forall$（任意）、$\exists$（存在）、$\Rightarrow$（推出）的顺序。

第三：解释每一次定理调用。从有界得到收敛子列还缺 Bolzano–Weierstrass；得到极限在原集合还需要闭性。

第四：检查结论是否恰好是题目所求。“有极限”不等于“极限属于集合”，“函数有界”不等于“达到最大值”。

最常用的三种模板

集合相等：任取 x，证明 $x \in A \Rightarrow x \in B$，再证明 $x \in B \Rightarrow x \in A$。

极限：给定任意 $\varepsilon > 0$，找出一个 N（或 δ）并验证最终不等式。

存在性：常用反证、上确界构造、嵌套区间、连续函数极值、固定点定理；对每个方法要明确其全部前提。

上半连续与下半连续先分清

上半连续（u.s.c.）关心：当 x 靠近 x_0 时，$\Phi(x)$ 不会逃出包住 $\Phi(x_0)$ 的开集。下半连续（l.s.c.）关心：$\Phi(x_0)$ 中每个点，在附近 $\Phi(x)$ 中仍能被逼近。Berge 最大值定理同时依赖两种方向，所以上下半连续都要会证明。

2 逐页课件与中文讲解

2.1 第 1 页｜本页课件

Theory of Value

An Axiomatic Analysis of Economic Equilibrium

Gerard Debreu (1959)

Professor Shunming Zhang

School of Finance
Renmin University of China

October 1, 2022

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

结合原文理解：读证明先辨认值集是否非空、紧、闭，以及定义域和目标空间的拓扑；缺任何一个假设都可能令结论不成立。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.2 第 2 页 | Chapter 1: Mathematics

Chapter 1: Mathematics

01. Introduction
02. Sets
03. Functions and Correspondences
04. Preorderings
05. Real Numbers
06. Limits in $\mathbb{R}^N$
07. Continuous Functions
08. Continuous Correspondences
09. Vectors in $\mathbb{R}^N$
10. Fixed Points

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 反身性是 xRx；传递性是 xRy 且 yRz　xRz。完全性与反对称性不包含在预序定义内。

读课件时的动作： 本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.3 第 3 页｜Continuous Correspondences

Continuous Correspondences

a. Example (2) of 1.3.h has pointed out the interest of correspondences for economics; this section will study their continuity.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

结合原文理解：读证明先辨认值集是否非空、紧、闭，以及定义域和目标空间的拓扑；缺任何一个假设都可能令结论不成立。

读课件时的动作：本页有例子或构造。自己换一个简单数值例子，算出具体集合或函数值；再思考一旦撤掉某条假设，结论是否还成立。

2.4 第 4 页｜Continuous Correspondences

Continuous Correspondences

b. In this section, S denotes a subset of $\mathbb{R}^K$, T a **compact subset** of $\mathbb{R}^L$, (x^q) a sequence of points of S, (y^q) a sequence of points of T, and a φ correspondence from S to T.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：在有限维欧氏空间，紧致 $\Leftrightarrow$ 闭且有界 $\Leftrightarrow$ 每个序列有收敛子列且极限仍在集合内。

读课件时的动作：本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.5 第 5 页｜Continuous Correspondences

Continuous Correspondences

c. The concept of continuity for a correspondence will be introduced in three steps.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

结合原文理解：读证明先辨认值集是否非空、紧、闭，以及定义域和目标空间的拓扑；缺任何一个假设都可能令结论不成立。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.6 第 6 页｜Continuous Correspondences

Continuous Correspondences

d. Let x^0 be a point of S; one defines firstly:
The correspondence φ is upper semicontinuous (u.s.c.) at the point x^0 if:

$$\text{“}x^q \to x^0, y^q \in \varphi\left(x^q\right), y^q \to y^0\text{”} \text{ implies } \text{“}y^0 \in \varphi\left(x^0\right).\text{”}$$

In other words, if x^q tends to x^0, and if y^q tends to y^0 while belonging for all q to the image-set of x^q, one requires that y^0 belong to the image-set of x^0. One could also say, if $x^q \to x^0$, and if the distance from y^0 to the image-set of x^q tends to zero, one requires that y^0 belong to the image-set of x^0. This is a natural but rather weak continuity requirement, as the following example shows. Let S and T be two compact real intervals. The graph of φ (fig. 2.a) is the shaded region, heavy-lined boundary included; $\varphi\left(x^0\right)$ is the interval $\left[a^1, a^2\right]$. The correspondence φ is closed at x^0.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：在有限维欧氏空间，紧致 $\Leftrightarrow$ 闭且有界 $\Leftrightarrow$ 每个序列有收敛子列且极限仍在集合内。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.7 第 7 页｜Continuous Correspondences

Continuous Correspondences

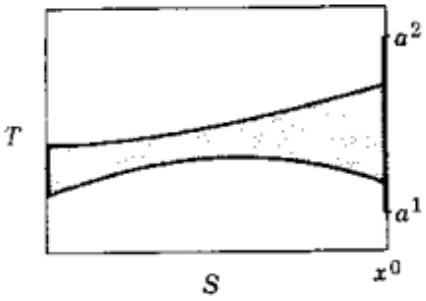


Fig. 2.a

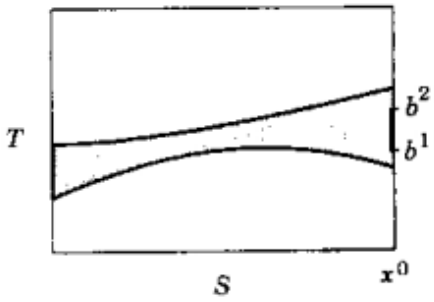


Fig. 2.b

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

结合原文理解：读证明先辨认值集是否非空、紧、闭，以及定义域和目标空间的拓扑；缺任何一个假设都可能令结论不成立。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论?

2.8 第 8 页｜Continuous Correspondences

Continuous Correspondences

e. One defines secondly:
The correspondence φ is lower semicontinuous (l.s.c.) at the point x^0 if:

"$x^q \to x^0, y^0 \in \varphi\left(x^0\right)$" implies "there is $\left(y^q\right)$ such that $y^q \in \varphi\left(x^q\right), y^q \to y^0$.

In other words, if x^q tends to x^0, and if y^0 belongs to the image-set of x^0, one requires that there be an infinite sequence $\left(y^q\right)$ such that y^q tends to y^0 while belonging for all q to the image-set of x^q. One could also say, if x^q tends to x^0, and if y^0 belongs to the image-set of x^0, one requires that the distance from y^0 to the image-set of x^q tend to zero. Again, this is a natural but weak continuity requirement as the example of fig. 2.b shows. Here $\varphi\left(x^0\right)$ is the interval $\left[b^1, b^2\right]$. The correspondence φ is lower semi-continuous at x_0. Notice how the roles of "$y^0 \in \varphi\left(x^0\right)$" and "$y^q \in \varphi\left(x^q\right)$, $y^q \to y^0$" are permuted in the two preceding definitions.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：下半连续要求：若 y Φ(x)，当自变量序列趋近给定点时，可以从临近的对应值集中挑选一列点逼近原来的目标点（在适当的度量空间设定下）。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

对应的完整证明：参见本讲义最后的「lsc」证明专题。

2.9 第 9 页 | Continuous Correspondences

Continuous Correspondences

f. Finally one defines:

The correspondence φ is continuous at the point x^0 if it is upper and lower semicontinuous at x^0.

φ then has all the desirable continuity properties at x^0.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：下半连续要求：若 y Φ(x)，当自变量序列趋近给定点时，可以从临近的对应值集中挑选一列点逼近原来的目标点（在适当的度量空间设定下）。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论?

对应的完整证明：参见本讲义最后的「lsc」证明专题。

2.10 第 10 页｜Continuous Correspondences

Continuous Correspondences

g. When, for all $x \in S$, $\varphi(x)$ consists of a single element, the definition of lower semicontinuity at x^0 is obviously equivalent to the definition of continuity at x^0 for a function. One can prove that, in the same case, the definition of upper semicontinuity at x^0 is equivalent to the definition of continuity at x^0 for a function.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

结合原文理解：读证明先辨认值集是否非空、紧、闭，以及定义域和目标空间的拓扑；缺任何一个假设都可能令结论不成立。

读课件时的动作：本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.11 第 11 页｜Continuous Correspondences

Continuous Correspondences

h. Semicontinuities and continuity on S are defined as semicontinuities and continuity at every point of S. It is clear that:

(1) The correspondence φ is upper semicontinuous on S if and only if its graph is closed in $S \times T$.

Proof. If the graph $G(\varphi)$ is closed in $S \times T$, then the correspondence φ is upper semicontinuous on S. For any $x^0 \in S$, if $x^q \to x^0$, $y^q \in \varphi(x^q)$, and $y^q \to y^0$, then $(x^q, y^q) \in G(\varphi)$ and $(x^q, y^q) \to (x^0, y^0)$. As $G(\varphi)$ is closed, then $(x^0, y^0) \in G(\varphi)$ and hence $y^0 \in \varphi(x^0)$.

Proof. If the correspondence φ is upper semicontinuous on S, then its graph is closed in $S \times T$. If $(x^q, y^q) \in G(\varphi)$ and $(x^q, y^q) \to (x^0, y^0)$, then $x^q \to x^0$, $y^q \to y^0$, and $y^q \in \varphi(x^q)$. As φ is upper semicontinuous on x^0, then $y^0 \in \varphi(x^0)$ and hence $(x^0, y^0) \in G(\varphi)$. □

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：上半连续对应可用“闭集上逆像为闭集”或序列判别，在紧值等假设下成立。不要把闭图像直接当成普遍等价条件。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

对应的完整证明：参见本讲义最后的「usc」证明专题。

2.12 第 12 页｜本页课件

Continuous Correspondences

i. As in 1.7.c let S_1, S_2, T be subsets of $\mathbb{R}^{K_1}$, $\mathbb{R}^{K_2}$, $\mathbb{R}^{L}$ respectively, f be a function from S_1 to S_2, and φ be a correspondence from S_2 to T. Define a correspondence ψ from S_1 to T by $\psi(x) = \varphi(f(x))$ for every x in S_1. It is immediate that:

(2) If f is continuous at the point x of S_1, and φ is upper semicontinuous (resp. lower semicontinuous) at the point $f(x)$ of S_2, then ψ is upper semicontinuous (resp. lower semicontinuous) at x.

Proof. If $x^q \to x$, $y^q \in \psi(x^q)$, and $y^q \to y$, as$f$ is continuous at the point x, then $z^q \equiv f(x^q) \to f(x) \equiv z$,
$y^q \in \psi(x^q) = \varphi(f(x^q)) \equiv \varphi(z^q)$. As φ is upper semicontinuous at the point $z \equiv f(x)$, then $y \in \varphi(z) = \varphi(f(x)) = \psi(x)$. Hence ψ is upper semicontinuous at x.
Proof. If $x^q \to x$, and $y \in \varphi(x)$, as$f$ is continuous at the point x, then $z^q \equiv f(x^q) \to f(x) \equiv z$, $y \in \psi(x) = \varphi(f(x)) \equiv \varphi(z)$. As φ is lower semicontinuous at the point $z \equiv f(x)$, then there is (y^q) such that $y^q \in \varphi(z^q) \equiv \varphi(f(x^q)) = \psi(x^q)$ and $y^q \to y$. Hence ψ is lower semicontinuous at x. □

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：上半连续对应可用“闭集上逆像为闭集”或序列判别，在紧值等假设下成立。不要把闭图像直接当成普遍等价条件。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

对应的完整证明：参见本讲义最后的「usc」证明专题。

2.13 第 13 页｜Continuous Correspondences

Continuous Correspondences

j. As in 1.7.d-e, for every $n = 1, 2, \cdots, N$ let T_n be a subset of $\mathbb{R}^{L_n}$, and φ_n be a correspondence from S to T_n. Consider the product $T = \prod_{n=1}^{N} T_n$ and define a correspondence φ from S to T by $\varphi(x) = \prod_{n=1}^{N} \varphi_n(x)$ for every x in S. If every T_n is compact, so is T by (7) of 1.6.t. In this case it is immediate that:

(3) If every φ_n is upper semicontinuous (resp. lower semicontinuous) at x of S, then φ is upper semicontinuous (resp. lower semicontinuous) at x.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：在有限维欧氏空间，紧致 ⇔ 闭且有界 ⇔ 每个序列有收敛子列且极限仍在集合内。

读课件时的动作：本页有例子或构造。自己换一个简单数值例子，算出具体集合或函数值；再思考一旦撤掉某条假设，结论是否还成立。

2.14 第 14 页｜ Continuous CorrespondencesProof. If xq →x, y q φ (xq),

Continuous Correspondences

Proof. If $x^q \to x$, $y^q \in \varphi(x^q)$, and $y^q \to y$, then $(y_1^q, \cdots, y_N^q) = y^q \in \varphi(x^q) = \prod_{n=1}^{N} \varphi_n(x^q)$ and $(y_1^q, \cdots, y_N^q) = y^q \to y = (y_1, \cdots, y_N)$, that is, $y_n^q \in \varphi_n(x^q)$ and $y_n^q \to y_n$ for $n = 1, \cdots, N$. As φ_n is upper semicontinuous at x, then $y_n \in \varphi_n(x)$ for $n = 1, \cdots, N$. Therefore $y = (y_1, \cdots, y_N) \in \prod_{n=1}^{N} \varphi_n(x) = \varphi(x)$. Hence φ is upper semicontinuous at x.

Proof. If $x^q \to x$, and $y \in \varphi(x)$, then $(y_1, \cdots, y_N) = y \in \varphi(x) = \prod_{n=1}^{N} \varphi_n(x)$, that is, $y_n \in \varphi_n(x)$ for $n = 1, \cdots, N$. As φ_n is lower semicontinuous at the point x, then there is (y_n^q) such that $y_n^q \in \varphi_n(z^q)$ and $y_n^q \to y_n$ for $n = 1, \cdots, N$. Thus $y^q = (y_1^q, \cdots, y_N^q) \in \prod_{n=1}^{N} \varphi_n(x^q) = \varphi(x^q)$ and $y^q = (y_1^q, \cdots, y_N^q) \to (y_1, \cdots, y_N) = y$. Hence φ is lower semicontinuous at x. □

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：上半连续对应可用“闭集上逆像为闭集”或序列判别，在紧值等假设下成立。不要把闭图像直接当成普遍等价条件。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

对应的完整证明：参见本讲义最后的「usc」证明专题。

2.15 第 15 页｜Continuous Correspondences

Continuous Correspondences

k. According to example (2) of 1.3.h. the interest of these concepts for economics lies, in particular, in the interpretations of an element x of S as the environment of a certain agent, of T as the set of actions a priori available to him, and of $\varphi(x)$ (assumed here to be closed for every x in S) as the subset of T to which his choice is actually restricted by his environment x. Let f be a continuous real-valued function on $S \times T$, and interpret $f(x, y)$ as the gain for that agent when his environment is x and his action y. Given x, one is interested in the elements of $\varphi(x)$ which maximize f (now a function of y alone) on $\varphi(x)$; they form a set $\mu(x)$. What can be said about the continuity of the correspondence μ from S to T?

$$\mu(x) = \{y \in \varphi(x) \mid y \text{ maximizes } f(x, y)\}$$

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

结合原文理解：读证明先辨认值集是否非空、紧、闭，以及定义域和目标空间的拓扑；缺任何一个假设都可能令结论不成立。

读课件时的动作：本页有例子或构造。自己换一个简单数值例子，算出具体集合或函数值；再思考一旦撤掉某条假设，结论是否还成立。

2.16 第 16 页｜ Continuous Correspondences

Continuous Correspondences

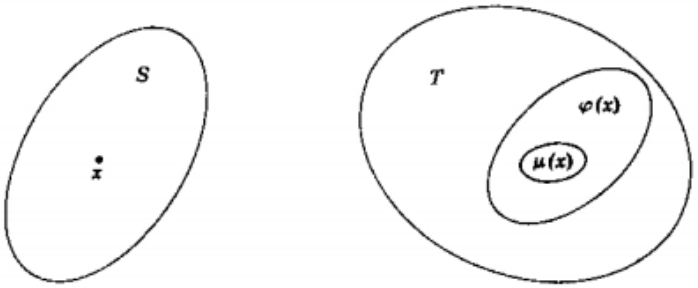


Fig. 3

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

结合原文理解：读证明先辨认值集是否非空、紧、闭，以及定义域和目标空间的拓扑；缺任何一个假设都可能令结论不成立。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.17 第 17 页｜Continuous Correspondences

Continuous Correspondences

One is also interested in $g(x)$, the value of the maximum of f on $\varphi(x)$ for a given x.

$$g(x) = \max\{f(x,y) \mid y \in \varphi(x)\}$$

What can be said about the continuity of the real-valued function g on S? An answer to these two questions is given by the following result (the proof of continuity of g should not be attempted).

(4) If f is continuous on $S \times T$, and if φ is continuous at $x \in S$, then μ is upper semicontinuous at x, and g is continuous at x.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：上半连续对应可用“闭集上逆像为闭集”或序列判别，在紧值等假设下成立。不要把闭图像直接当成普遍等价条件。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

对应的完整证明：参见本讲义最后的「usc」证明专题。

2.18 第 18 页｜ Continuous Correspondences

Continuous Correspondences

(4) If f is continuous on $S \times T$, and if φ is compact-valued continuous at $x_0 \in S$, then μ is upper semicontinuous at x_0, and g is continuous at x_0.

(4) If f is continuous on $S \times T$, and if φ is compact-valued (lower semi-continuous and upper semi-continuous) continuous at $x_0 \in S$, define the optimal value function $g : S \to \mathbb{R}$,

$$g(x) = \max \{f(x, y) \mid y \in \varphi(x)\},$$

and argmax correspondence $\mu : S \to T$,

$$\begin{aligned} \mu(x) &= \{y \in \varphi(x) \mid y \text{ maximizes } f(x, y)\} \\ &= \{y \in \varphi(x) \mid f(x, y) = g(x)\} \end{aligned}$$

then g is continuous at x_0, and μ is compact-valued upper semicontinuous at x_0.

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 在有限维欧氏空间，紧致 $\Leftrightarrow$ 闭且有界 $\Leftrightarrow$ 每个序列有收敛子列且极限仍在集合内。

读课件时的动作： 本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.19 第 19 页｜本页课件

Continuous Correspondences

Proof. Fix $x_0 \in S$ and let $x_n \to x_0$. Since φ is lower semi-continuous continuous (l.s.c.) at $x_0 \in S$, for any $y_0 \in \varphi(x_0)$, there exists $(y_n) \subseteq T$ such that $y_n \in \varphi(x_n)$ and $y_n \to y^0$. By the continuity of f, we have $f(x_n, y_n) \to f(x_0, y_0)$. Since $f(x_n, y_n) \leqq g(x_n)$, then

$$f(x_0, y_0) = \lim_{n\to\infty} f(x_n, y_n) \leqq \underline{\lim}_{n\to\infty} g(x_n)$$

and hence $g(x_0) \leqq \underline{\lim}_{n\to\infty} g(x_n)$. Therefore g is lower semi-continuous continuous (l.s.c.) at x_0.

Fix $x_0 \in S$ and let $x_n \to x_0$. Since φ is compact-valued upper semi-continuous continuous (u.s.c.) at $x_0 \in S$, for any sequence $(y_n \in \varphi(x_n))$, the exists a subsequence $(y_{n_k} \in \varphi(x_{n_k}))$ such that $y_{n_k} \to y_0$ and $y^0 \in \varphi(x_0)$. By the continuity of f, we have $f(x_{n_k}, y_{n_k}) \to f(x_0, y_0)$. Since $f(x_{n_k}, y_{n_k}) = g(x_{n_k})$, then

$$g(x_0) = f(x_0, y_0) = \lim_{n\to\infty} f(x_n, y_n) \geqq \overline{\lim}_{n\to\infty} g(x_n).$$

Therefore g is upper semi-continuous continuous (u.s.c.) at x_0.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：在有限维欧氏空间，紧致 $\Leftrightarrow$ 闭且有界 $\Leftrightarrow$ 每个序列有收敛子列且极限仍在集合内。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.20 第 20 页｜ Continuous Correspondences

Continuous Correspondences

Proof. For each x_0, f is continuous on $S \times T$, and if φ is compact-valued continuous at $x_0 \in S$, then $\mu(x_0)$ is non-empty and compact-valued from the Extreme Value Theorem.

Let $x_n \to x_0$, $y_n \in \mu(x_n)$. Since φ is compact-valued upper semi-continuous continuous (u.s.c.) at $x_0 \in S$, then there exists $(y_{n_k}) \subseteq T$ such that $y_{n_k} \in \varphi(x_{n_k})$ and $y_{n_k} \to y_0$ and $y_0 \in \varphi(x_0)$. By the continuity of f, we have $f(x_{n_k}, y_{n_k}) \to f(x_0, y_0)$. By the continuity of g, we have $g(x_{n_k}) \to g(x_0)$. Since $f(x_{n_k}, y_{n_k}) = g(x_{n_k})$, then $f(x_0, y_0) = g(x_0)$, so $y_0 \in \mu(x_0)$. Thus μ is upper semicontinuous (u.s.c.) at x_0. □

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 紧集上的连续函数必有最大值与最小值；“有界”和“取到界”是两个不同的命题。

读课件时的动作： 本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

对应的完整证明： 参见本讲义最后的「evt」证明专题。

2.21 第 21 页｜Continuous Correspondences

Continuous Correspondences

Example: Let S and T be the real interval $[0,1]$. Define φ by $\varphi(x) = [0,1]$ for all $x \in S$, and f by $f(x,y) = xy$. For $x \neq 0$, $\mu(x)$ consists of the single element 1; for $x = 0$, $\mu(x) = [0,1]$.

$$\mu(x) = \begin{cases} [0,1], & \text{for} \quad x = 0, \\ \{1\}, & \text{for} \quad x \neq 0. \end{cases}$$

(It will be helpful to draw the graph of μ.) For all x, $g(x) = x$.

$$g(x) = \max\{xy \mid y \in [0,1]\} = x \max\{y \mid y \in [0,1]\} = x.$$

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

结合原文理解：读证明先辨认值集是否非空、紧、闭，以及定义域和目标空间的拓扑；缺任何一个假设都可能令结论不成立。

读课件时的动作：本页有例子或构造。自己换一个简单数值例子，算出具体集合或函数值；再思考一旦撤掉某条假设，结论是否还成立。

2.22 第 22 页｜Continuous Correspondences

Continuous Correspondences

I. Throughout this section, the assumption that T is compact has been made; it is essential in several respects. In applications, when the set T is not compact, one may still be able to replace T by some compact set without changing the problem, and thus to use the above results.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：在有限维欧氏空间，紧致 $\Leftrightarrow$ 闭且有界 $\Leftrightarrow$ 每个序列有收敛子列且极限仍在集合内。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.23 第 23 页｜本页课件

Semi-continuous Mappings

In this section, S denotes a subset of $\mathbb{R}^K$, T a subset of $\mathbb{R}^L$, and a φ correspondence from S to T, that is $\varphi : S \rightrightarrows T$. For correspondences there are two natural notions of inverse.

(1) The lower (or weak) inverse of $E \subseteq T$ under $\varphi : S \rightrightarrows T$, denoted $\varphi^-(E)$, is defined by

$$\varphi^-(E) = \{x \in S : \varphi(x) \cap E \neq \varnothing\}.$$

(2) The upper (or strong) inverse of $E \subseteq T$ under $\varphi : S \rightrightarrows T$, denoted $\varphi^+(E)$, is defined by

$$\varphi^+(E) = \{x \in S : \varphi(x) \subseteq E\}.$$

Remark. For a single-valued mapping $\varphi : S \rightrightarrows T$, define

$$\varphi^{-1}(y) = \{x \in S : y \in \varphi(x)\} = \{x \in S : y = \varphi(x)\}.$$

Note that $\varphi^{-1}(y) = \varphi^-(\{y\})$.

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

结合原文理解： 读证明先辨认值集是否非空、紧、闭，以及定义域和目标空间的拓扑；缺任何一个假设都可能令结论不成立。

读课件时的动作： 本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.24 第 24 页｜本页课件

Semi-continuous Mappings

Let φ be a correspondence of a topological space S into a topological space T and let x_0 be a point of S.

- We say that φ is lower semi-continuous (l.s.c.) at x_0 if for each open set G meeting $\varphi(x_0)$ $[\varphi(x_0) \cap G \neq \varnothing]$ there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \cap G \neq \varnothing,$$

  i.e. $\varphi(U(x_0)) \cap G \neq \varnothing$.

- We say that φ is upper semi-continuous (u.s.c.) at x_0 if for each open set G containing $\varphi(x_0)$ $[\varphi(x_0) \subseteq G]$ there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \subseteq G,$$

  i.e. $\varphi(U(x_0)) \subseteq G$.

We say that the mapping φ is continuous at x_0 if it is both lower and upper semi-continuous at x_0.

$$\varphi^-(E) = \{x \in S \mid \varphi(x) \cap E \neq \varnothing\}$$
$$\varphi^+(E) = \{x \in S \mid \varphi(x) \subseteq E\}$$

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.25 第 25 页｜ Semi-continuous Mappings

Semi-continuous Mappings

- We say that φ is lower semi-continuous (l.s.c.) at x_0 if for each open set G meeting $\varphi(x_0)$ $[\varphi(x_0) \cap G \neq \varnothing]$ there exists a neighbourhood $U(x_0)$ such that $\varphi(U(x_0)) \cap G \neq \varnothing$

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \cap G \neq \varnothing.$$

- We say that φ is lower semi-continuous (l.s.c.) at x_0 if for each open set G meeting $\varphi(x_0)$ $[x_0 \in \varphi^-(G)]$ there exists a neighbourhood $U(x_0)$ such that $U(x_0) \subseteq \varphi^-(G)$

$$x \in U(x_0) \quad \Longrightarrow \quad x \in \varphi^-(G).$$

- We say that φ is lower semi-continuous (l.s.c.) at x_0 if for each open set G meeting $\varphi(x_0)$ $[x_0 \in \varphi^-(G)]$ there exists a neighbourhood $U(x_0)$ such that $U(x_0) \subseteq \varphi^-(G)$.
- We say that φ is lower semi-continuous (l.s.c.) at x_0 if $\varphi^-(G)$ is a open set in S for each open set G (meeting $\varphi(x_0)$).

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.26 第 26 页｜Semi-continuous Mappings

Semi-continuous Mappings

- We say that φ is upper semi-continuous (u.s.c.) at x_0 if for each open set G containing $\varphi(x_0)$ $[\varphi(x_0) \subseteq G]$ there exists a neighbourhood $U(x_0)$ such that $\varphi(U(x_0)) \subseteq G$

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \subseteq G.$$

- We say that φ is upper semi-continuous (u.s.c.) at x_0 if for each open set G containing $\varphi(x_0)$ $[x_0 \in \varphi^+(G)]$ there exists a neighbourhood $U(x_0)$ such that $U(x_0) \subseteq \varphi^+(G)$

$$x \in U(x_0) \quad \Longrightarrow \quad x \in \varphi^+(G).$$

- We say that φ is upper semi-continuous (u.s.c.) at x_0 if for each open set G containing $\varphi(x_0)$ $[x_0 \in \varphi^+(G)]$ there exists a neighbourhood $U(x_0)$ such that $U(x_0) \subseteq \varphi^+(G)$.
- We say that φ is upper semi-continuous (u.s.c.) at x_0 if $\varphi^+(G)$ is a open set in S for each open set G (containing $\varphi(x_0)$).

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.27 第 27 页｜Semi-continuous Mappings

Semi-continuous Mappings

If φ is a single-valued mapping, the definition given above for lower semi-continuity coincides with the ordinary definition of continuity; the same is true for upper semi-continuity.

- We say that φ is lower semi-continuous (l.s.c.) in S if it is lower semi-continuous (l.s.c.) at each point of S, that is, $\varphi^{-}(G)$ is a open set in S for each open set G.
- We say that φ is upper semi-continuous (u.s.c.) in S if it is upper semi-continuous (u.s.c.) at each point of S, that is, $\varphi^{+}(G)$ is a open set in S for each open set G.

We say that the correspondence φ is continuous in S if it is both lower and upper semi-continuous in S.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作：本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.28 第 28 页｜ Semi-continuous Mappings

Semi-continuous Mappings

Theorem 1.1. $\varphi : S \rightrightarrows T$ is a correspondence, then the following three statements are equivalent.

(1) φ is lower semi-continuous (l.s.c.).

(2) $\varphi^{-}(G)$ is a open set in S for any open set $G \subseteq T$.

(3) $\varphi^{+}(F)$ is a closed set in S for any closed set $F \subseteq T$.

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作： 本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

2.29 第 29 页｜本页课件

Semi-continuous Mappings

Proof of (1) $\Longrightarrow$ (2) For any open set $G \subseteq T$ we want to prove that $\varphi^{-}(G)$ is a open set in S. If $x_0 \in \varphi^{-}(G)$, then $\varphi(x_0) \cap G \neq \varnothing$, and so there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \cap G \neq \varnothing,$$

i.e. $\varphi(U(x_0)) \cap G \neq \varnothing$, whence

$$x \in U(x_0) \quad \Longrightarrow \quad x \in \varphi^{-}(G),$$

i.e. $U(x_0) \subseteq \varphi^{-}(G)$. Therefore $\varphi^{-}(G)$ is open.

Proof of (2) $\Longrightarrow$ (1) Suppose now that $\varphi^{-}(G)$ is an open set for each open set G in T. If $x_0 \in \varphi^{-}(G)$, then there exists a neighbourhood $U(x_0)$ such that $U(x_0) \subseteq \varphi^{-}(G)$, that is, if G meets $\varphi(x_0)$ $[\varphi(x_0) \cap G \neq \varnothing]$, then there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad x \in \varphi^{-}(G),$$

hence

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \cap G \neq \varnothing,$$

i.e. $\varphi(U(x_0)) \cap G \neq \varnothing$, thus φ is lower semi-continuous (l.s.c.) at x_0.

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作： 本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.30 第 30 页｜ Semi-continuous MappingsProof of (2) = (3) For any clos

Semi-continuous Mappings

Proof of (2) $\Longrightarrow$ (3) For any closed set $F \subseteq T$, $G = \mathbf{C}_T F$ is an open set.

$$\begin{aligned}
\varphi^-(G) &= \{x \in S \mid \varphi(x) \cap G \neq \varnothing\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \cap G = \varnothing\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \subseteq \mathbf{C}_T G\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \subseteq F\} \\
&= \mathbf{C}_S\left(\varphi^+(F)\right)
\end{aligned}$$

is open, that is, $\varphi^+(F)$ is closed.

Proof of (3) $\Longrightarrow$ (2) For any open set $G \subseteq T$, $F = \mathbf{C}_T G$ is a closed set.

$$\begin{aligned}
\varphi^+(F) &= \{x \in S \mid \varphi(x) \subseteq F\} \\
&= \{x \in S \mid \varphi(x) \subseteq \mathbf{C}_T G\} \\
&= \{x \in S \mid \varphi(x) \cap G = \varnothing\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \cap G \neq \varnothing\} \\
&= \mathbf{C}_S\left(\varphi^-(G)\right)
\end{aligned}$$

is closed, that is, $\varphi^-(G)$ is open. □

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作： 本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.31 第 31 页｜ Semi-continuous Mappings

Semi-continuous Mappings

Theorem 1.2. $\varphi : S \rightrightarrows T$ is a correspondence, then the following three statements are equivalent.

(1) φ is upper semi-continuous (u.s.c.).

(2) $\varphi^+(G)$ is a open set in S for any open set $G \subseteq T$.

(3) $\varphi^-(F)$ is a closed set in S for any closed set $F \subseteq T$.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

2.32 第 32 页｜本页课件

Semi-continuous Mappings

Proof of (1) $\Longrightarrow$ (2) For any open set $G \subseteq T$ we want to prove that $\varphi^+(G)$ is a open set in S. If $x_0 \in \varphi^+(G)$, then $\varphi(x_0) \subseteq G$, and so there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \subseteq G,$$

i.e. $\varphi(U(x_0)) \subseteq G$, whence

$$x \in U(x_0) \quad \Longrightarrow \quad x \in \varphi^+(G),$$

i.e. $U(x_0) \subseteq \varphi^+(G)$. Therefore $\varphi^+(G)$ is a neighbourhood of each of its points and so is open.

Proof of (2) $\Longrightarrow$ (1) Suppose now that $\varphi^+(G)$ is an open set for each open set G in T. If $x_0 \in \varphi^+(G)$, then there exists a neighbourhood $U(x_0)$ such that $U(x_0) \subseteq \varphi^+(G)$, that is, if G contains $\varphi(x_0)$ $[\varphi(x_0) \subseteq G]$, then there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad x \in \varphi^+(G),$$

hence

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \subseteq G,$$

i.e. $\varphi(U(x_0)) \subseteq G$, thus φ is upper semi-continuous (u.s.c.) at x_0.

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作： 本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.33 第 33 页｜ Semi-continuous MappingsProof of (2) = (3) For any clos

Semi-continuous Mappings

Proof of (2) $\Longrightarrow$ (3) For any closed set $F \subseteq T$, $G = \mathbf{C}_T F$ is an open set.

$$\begin{aligned}
\varphi^+(G) &= \{x \in S \mid \varphi(x) \subseteq G\} \\
&= \{x \in S \mid \varphi(x) \subseteq \mathbf{C}_T F\} \\
&= \{x \in S \mid \varphi(x) \cap F = \varnothing\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \cap F \neq \varnothing\} \\
&= \mathbf{C}_S\left(\varphi^-(F)\right)
\end{aligned}$$

is open, that is, $\varphi^-(F)$ is closed.

Proof of (3) $\Longrightarrow$ (2) For any open set $G \subseteq T$, $F = \mathbf{C}_T G$ is a closed set.

$$\begin{aligned}
\varphi^-(F) &= \{x \in S \mid \varphi(x) \cap F \neq \varnothing\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \cap F = \varnothing\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \subseteq \mathbf{C}_T F\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \subseteq G\} \\
&= \mathbf{C}_S\left(\varphi^+(G)\right)
\end{aligned}$$

is closed, that is, $\varphi^+(G)$ is open. □

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作： 本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.34 第 34 页｜Semi-continuous Mappings

Semi-continuous Mappings

Upper semi-continuous (u.s.c.) correspondences of S into T have the following fundamental property:

Theorem 1.3. If $\varphi : S \to T$ is a compact-valued u.s.c. correspondence, then the image $\varphi(K)$ of a compact subset K of S is also compact.

Proof. Let $\{G_\lambda \mid \lambda \in \Lambda\}$ be an open covering of $\varphi(K)$. If $x \in K$, the set $\varphi(x)$, which is compact, can be covered by a finite number of the $G_\lambda(x)$ for $\lambda \in \Lambda$, $\varphi(x) \subseteq \bigcup_{\lambda\in\Lambda} G_\lambda(x)$; let G_x denote the union of the sets in such a finite family, $G_x = \bigcup_{\lambda\in\Lambda} G_\lambda(x)$, then $\varphi(K) \subseteq \bigcup_{x\in K} \varphi(x) \subseteq \bigcup_{x\in K} G_x$. Then $\{\varphi^+(G_x) \mid x \in K\}$ is an open covering of K and so it contains a finite covering $\varphi^+(G_{x_1}), \cdots, \varphi^+(G_{x_n}), \cdots, \varphi^+(G_{x_N})$. $K \subseteq \bigcup_{n=1}^{N} \varphi^+(G_{x_n})$. The sets $G_{x_1}, \cdots, G_{x_n}, \cdots, G_{x_N}$ cover $\varphi(K)$ and so $\varphi(K)$ can be covered by a finite number of the G_{x_n} for $n = 1, \cdots, N$. That is, $\varphi(K) \subseteq \bigcup_{n=1}^{N} G_{x_n}$.
□

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：开覆盖具有有限子覆盖的结论必须依赖紧致性。证明时可用闭区间套反证或逐次细分。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.35 第 35 页｜Semi-continuous Mappings

Semi-continuous Mappings

In addition to the two types of semi-continuity, it is sometimes convenient to consider a third topological property.

Definition 1.2. (Closed Correspondence) We say that φ is a closed correspondence of S into T if whenever $x_0 \in S$, $y_0 \in T$, $y_0 \notin \varphi(x_0)$ there exist two neighbourhoods $U(x_0)$ and $V(y_0)$ such that $x \in U(x_0)$ implies $\varphi(x) \cap V(y_0) = \varnothing$.

Consider the graphical representation $\sum_{x \in S} \varphi(x)$ of φ in $S \times T$;

$$G(\varphi) \equiv \sum_{x \in S} \varphi(x) = \{(x, y) \in S \times T \mid y \in \varphi(x)\}.$$

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

结合原文理解：读证明先辨认值集是否非空、紧、闭，以及定义域和目标空间的拓扑；缺任何一个假设都可能令结论不成立。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.36 第 36 页｜Semi-continuous Mappings

Semi-continuous Mappings

- We say that φ is a closed correspondence of S into T if whenever $x_0 \in S$, $y_0 \in T$, $y_0 \notin \varphi(x_0)$ there exist two neighbourhoods $U(x_0)$ and $V(y_0)$ such that $x \in U(x_0)$ implies $\varphi(x) \cap V(y_0) = \varnothing$.
- We say that φ is a closed correspondence of S into T if whenever $x_0 \in S$, $y_0 \in T$, $y_0 \notin \varphi(x_0)$ there exist two neighbourhoods $U(x_0)$ and $V(y_0)$ such that $x \in U(x_0)$ and $y \in V(y_0)$ implies $y \notin \varphi(x)$.
- We say that φ is a closed correspondence of S into T if whenever $x_0 \in S$, $y_0 \in T$, $(x_0, y_0) \in \mathbf{C}(G(\varphi))$ there exist two neighbourhoods $U(x_0)$ and $V(y_0)$ such that $x \in U(x_0)$ and $y \in V(y_0)$ implies $(x, y) \in \mathbf{C}(G(\varphi))$.
- We say that φ is a closed correspondence of S into T if whenever $x_0 \in S$, $y_0 \in T$, $(x_0, y_0) \in \mathbf{C}(G(\varphi))$ there exist two neighbourhoods $U(x_0)$ and $V(y_0)$ such that $U(x_0) \times V(y_0) \subseteq \mathbf{C}(G(\varphi))$.
- We say that φ is a closed correspondence of S into T if $\mathbf{C}(G(\varphi))$ is an open set in $S \times T$.
- We say that φ is a closed correspondence of S into T if $G(\varphi)$ is a closed set in $S \times T$.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.37 第 37 页｜ Semi-continuous Mappings

Semi-continuous Mappings

Proposition 1.1. Let S and T be two metric spaces. A mapping $\varphi : S \rightrightarrows T$ is a closed correspondence if and only if $G(\varphi)$ is a closed set in $S \times T$.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

2.38 第 38 页｜Semi-continuous Mappings

Semi-continuous Mappings

- We say that φ is a closed correspondence of S into T if whenever $x_0 \in S$, $y_0 \in T$, $y_0 \notin \varphi(x_0)$ there exist two neighbourhoods $U(x_0)$ and $V(y_0)$ such that $x \in U(x_0)$ implies $\varphi(x) \cap V(y_0) = \varnothing$.
- We say that φ is a closed correspondence of S into T if whenever $x_0 \in S$, $y_0 \in T$, for any two neighbourhoods $U(x_0)$ and $V(y_0)$ such that $x \in U(x_0)$ implies $\varphi(x) \cap V(y_0) \neq \varnothing$ [$y \in \varphi(x) \cap V(y_0)$ if and only if $y \in \varphi(x)$ and $y \in V(y_0)$], then $y_0 \in \varphi(x_0)$.
- We say that φ is a closed correspondence of S into T if whenever $x_0 \in S$, $y_0 \in T$, for any two neighbourhoods $U(x_0)$ and $V(y_0)$ such that $x \in U(x_0)$, $y \in V(y_0)$, and $y \in \varphi(x)$, then $y_0 \in \varphi(x_0)$.
- We say that φ is a closed correspondence of S into T if whenever $x_n \longrightarrow x_0$, $y_n \longrightarrow y_0$, $y_n \in \varphi(x_n)$, then $y_0 \in \varphi(x_0)$.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

结合原文理解：读证明先辨认值集是否非空、紧、闭，以及定义域和目标空间的拓扑；缺任何一个假设都可能令结论不成立。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.39 第 39 页｜Semi-continuous Mappings

Semi-continuous Mappings

Proposition 1.2. Let S and T be two metric spaces. A mapping $\varphi : S \rightrightarrows T$ is a closed correspondence if and only if whenever $x_n \longrightarrow x_0$, $y_n \longrightarrow y_0$, $y_n \in \varphi(x_n)$, then $y_0 \in \varphi(x_0)$.

Theorem 1.4. $G(\varphi)$ is a closed set in $S \times T$ if and only if whenever $x_n \longrightarrow x_0$, $y_n \longrightarrow y_0$, $y_n \in \varphi(x_n)$, then $y_0 \in \varphi(x_0)$.

Proof. $G(\varphi)$ is a closed set in $S \times T$ if and only if whenever $(x_n, y_n) \longrightarrow (x_0, y_0)$ and $(x_n, y_n) \in G(\varphi)$, then $(x_0, y_0) \in G(\varphi)$ if and only if whenever $x_n \longrightarrow x_0$, $y_n \longrightarrow y_0$, $y_n \in \varphi(x_n)$, then $y_0 \in \varphi(x_0)$. □

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.40 第 40 页｜本页课件

Properties of Semi-continuous Mappings

Proposition 1.3. If every $\varphi_n : S \rightrightarrows T_n$ is a closed correspondence for $n = 1, \cdots, N$, then the Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n : S \rightrightarrows \prod_{n=1}^{N} T_n$ is a closed correspondence.

Proof 1. If $\left(y_1^0, \cdots, y_N^0\right) = y^0 \notin \varphi\left(x^0\right) = \prod_{n=1}^{N} \varphi_n\left(x^0\right)$, then there exists $n_0 = 1, \cdots, N$ such that $y_{n_0}^0 \notin \varphi_{n_0}\left(x^0\right)$. Therefore there exist neighbourhoods $U_{n_0}\left(x^0\right)$ and $V_{n_0}\left(y_{n_0}^0\right)$ such that $x \in U_{n_0}\left(x^0\right)$ implies $\varphi_{n_0}(x) \cap V_{n_0}\left(y_{n_0}^0\right) = \varnothing$, that is, $\varphi_{n_0}\left(U_{n_0}\left(x^0\right)\right) \cap V_{n_0}\left(y_{n_0}^0\right) = \varnothing$. Take $U\left(x^0\right) = \bigcap_{n=1}^{N} U_n\left(x^0\right)$, then $x \in U(x_0)$ implies

$$\varphi_{n_0}\left(U\left(x^0\right)\right) \cap V_{n_0}\left(y_{n_0}^0\right) \subseteq \varphi_{n_0}\left(U_{n_0}\left(x^0\right)\right) \cap V_{n_0}\left(y_{n_0}^0\right) = \varnothing$$

and hence $\varphi_{n_0}\left(U\left(x^0\right)\right) \cap V_{n_0}\left(y_{n_0}^0\right) = \varnothing$. Take $V\left(y^0\right) = \prod_{n=1}^{N} V_n\left(y_n^0\right)$, then

$$\varphi\left(U\left(x^0\right)\right) \cap V\left(y^0\right) = \prod_{n=1}^{N}\left[\varphi_n\left(U\left(x^0\right)\right) \cap V_n\left(y_n^0\right)\right] = \varnothing;$$

whence φ is a closed mapping. □

Proof 2. If $x^q \to x^0$, $\left(y_1^q, \cdots, y_N^q\right) = y^q \to y^0 = \left(y_1^0, \cdots, y_N^0\right)$, and $\left(y_1^q, \cdots, y_N^q\right) = y^q \in \varphi\left(x^q\right) = \prod_{n=1}^{N} \varphi_n\left(x^q\right)$, then $y_n^q \to y_n^0$ and $y_n^q \in \varphi_n\left(x^q\right)$ for $n = 1, \cdots, N$. If every φ_n is closed at the point x_0 of S, then $y_n^0 \in \varphi_n\left(x^0\right)$ for $n = 1, \cdots, N$, and hence $y^0 = \left(y_1^0, \cdots, y_N^0\right) \in \prod_{n=1}^{N} \varphi_n\left(x^0\right) = \varphi\left(x^0\right)$. Thus, $\varphi = \prod_{n=1}^{N} \varphi_n$ is closed at the point x_0 of S. □

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

结合原文理解： 读证明先辨认值集是否非空、紧、闭，以及定义域和目标空间的拓扑；缺任何一个假设都可能令结论不成立。

读课件时的动作： 本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.41 第 41 页 | Semi-continuous Mappings

Semi-continuous Mappings

Proposition 1.4. Let S and T be two metric spaces. If $\varphi : S \rightrightarrows T$ is a closed correspondence at x, then the set $\varphi(x)$ is closed in T.

Proof: Let y_n be a sequence in $\varphi(x)$ with $y_n \to y$. Define $x_n = x$ for all n, so $x_n \to x$ and $y_n \in \varphi(x_n)$. Since φ is a closed correspondence at x, we have $y \in \varphi(x)$. This shows that $\varphi(x)$ is a closed set. □

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作： 本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.42 第 42 页｜ Semi-continuous Mappings

Semi-continuous Mappings

Example 1.1. If f is a continuous numerical function in $S \times T$, the correspondence $\varphi : S \rightrightarrows T$ defined by $\varphi(x) = \{y \in T \mid f(x, y) \leqq 0\}$ is a closed correspondence of S into T, for, in $S \times T$, the graphical representation

$$G(\varphi) = \{(x, y) \in S \times T \mid f(x, y) \leqq 0\}$$

is a closed set.

In particular, if λ is a continuous numerical function in a metric space (X, d), the mapping

$$\varphi(x) = B_{\lambda(x)}(x) = \{y \in X \mid d(x, y) - \lambda(x) \leqq 0\}$$

is closed.

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作： 本页有例子或构造。自己换一个简单数值例子，算出具体集合或函数值；再思考一旦撤掉某条假设，结论是否还成立。

2.43 第 43 页｜ Semi-continuous MappingsTheorem 1.5. If {φλ | λ Λ} is

Semi-continuous Mappings

Theorem 1.5. If $\{\varphi_\lambda \mid \lambda \in \Lambda\}$ is a family of closed correspondences of S into T, then $\varphi = \bigcap_{\lambda\in\Lambda} \varphi_\lambda$ is also a closed correspondence.

Proof 1. If $y_0 \notin \varphi(x_0) = \bigcap_{\lambda\in\Lambda} \varphi_\lambda(x_0)$, then there exists an index $\lambda_0 \in$ such that

$$y_0 \notin \varphi_{\lambda_0}(x_0).$$

Therefore there exist neighbourhoods $U(x_0)$ and $V(y_0)$ such that $x \in U(x_0)$ implies $\varphi_{\lambda_0}(x) \cap V(y_0) = \varnothing$, that is,

$$\varphi_{\lambda_0}(U(x_0)) \cap V(y_0) = \varnothing$$

and so $x \in U(x_0)$ implies $\varphi(x) \cap V(y_0) = \bigcap_{\lambda\in\Lambda} [\varphi_\lambda(x) \cap V(y_0)] = \varnothing$, that is,

$$\varphi(U(x_0)) \cap V(y_0) = \varnothing;$$

whence φ is a closed mapping. □

Proof 2. If $x_n \to x_0$, $y_n \to y_0$, and $y_n \in \varphi(x_n) = \bigcap_{\lambda\in\Lambda} \varphi_\lambda(x_n)$, then $y_n \in \varphi_\lambda(x_n)$ for $\lambda \in \Lambda$. If every φ_λ is closed at the point x_0 of S, then $y_0 \in \varphi_\lambda(x_0)$ for $\lambda \in \Lambda$, and hence $y_0 \in \bigcap_{\lambda\in\Lambda} \varphi_\lambda(x_0) = \varphi(x_0)$. Thus, $\varphi = \bigcap_{\lambda\in\Lambda} \varphi_\lambda$ is closed at the point x_0 of S. □

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

结合原文理解：读证明先辨认值集是否非空、紧、闭，以及定义域和目标空间的拓扑；缺任何一个假设都可能令结论不成立。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.44 第 44 页｜ Properties of Semi-continuous Mappings

Properties of Semi-continuous Mappings

Proposition 1.5. If every $\varphi_n : S \rightrightarrows T_n$ is a closed correspondence for $n = 1, \cdots, N$, then the union $\varphi = \bigcup_{n=1}^{N} \varphi_n$ is a closed correspondence.

Proof 1. If $y_0 \notin \varphi(x_0) = \bigcup_{n=1}^{N} \varphi_n(x_0)$, then, for all $n = 1, \cdots, N$,

$$y_0 \notin \varphi_n(x_0).$$

Therefore there exist neighbourhoods $U(x_0)$ and $V(y_0)$ such that $x \in U(x_0)$ implies $\varphi_n(x) \cap V(y_0) = \varnothing$, that is,

$$\varphi_n(U(x_0)) \cap V(y_0) = \varnothing$$

and so $x \in U(x_0)$ implies $\varphi(x) \cap V(y_0) = \bigcup_{n=1}^{N} [\varphi_n(x) \cap V(y_0)] = \varnothing$, that is,

$$\varphi(U(x_0)) \cap V(y_0) = \bigcup_{n=1}^{N} [\varphi_n(U(x_0)) \cap V(y_0)] = \varnothing;$$

whence φ is a closed mapping. □

Proof 2. If $x^q \to x^0$, $y^q \to y^0$, and $y^q \in \varphi(x^q) = \bigcup_{n=1}^{N} \varphi_n(x^q)$, then there exists $n_0 \in \{1, \cdots, N\}$ such that $y^q \in \varphi_{n_0}(x^q)$. If every φ_n is closed at the point x_0 of S, then $y^0 \in \varphi_{n_0}\left(x^0\right)$, and hence $y^0 \in \bigcup_{n=1}^{N} \varphi_n\left(x^0\right) = \varphi\left(x^0\right)$. Thus, $\varphi = \bigcup_{n=1}^{N} \varphi_n$ is closed at the point x_0 of S. □

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

结合原文理解：读证明先辨认值集是否非空、紧、闭，以及定义域和目标空间的拓扑；缺任何一个假设都可能令结论不成立。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.45 第 45 页｜ Semi-continuous MappingsTheorem 1.6. Every compact-valu

Semi-continuous Mappings

Theorem 1.6. Every compact-valued u.s.c. correspondence is closed.

Proof. Let φ be a upper semi-continuous (u.s.c.) correspondence of S into T and suppose that $y_0 \notin \varphi(x_0)$. Since $\varphi(x_0)$ is compact, there exists an open set G in T containing $\varphi(x_0)$ and a neighbourhood $V(y_0)$ such that

$$G \cap V(y_0) = \varnothing.$$

Since φ is a upper semi-continuous (u.s.c.) correspondence, there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \Longrightarrow \varphi(x) \subseteq G.$$

Then

$$x \in U(x_0) \Longrightarrow \varphi(x) \cap V(y_0) = \varnothing$$

and φ is therefore closed. □

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作： 本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.46 第 46 页｜ Semi-continuous Mappings

Semi-continuous Mappings

Theorem 1.7. If φ_1 is a closed mapping of S into T and φ_2 is a compact-valued u.s.c. correspondence of S into T, then the correspondence $\varphi = \varphi_1 \cap \varphi_2$ is compact-valued u.s.c.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：在有限维欧氏空间，紧致 $\Leftrightarrow$ 闭且有界 $\Leftrightarrow$ 每个序列有收敛子列且极限仍在集合内。

读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

2.47 第 47 页｜本页课件

Semi-continuous Mappings

Proof. For each x, the set $\varphi(x)$ is compact, because it is a closed set contained in a compact set $\varphi_2(x)$. Thus φ is compact-valued.
Let G be an open set such that $\varphi(x_0) = \varphi_1(x_0) \cap \varphi_2(x_0) \subseteq G$. We shall prove that there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \Longrightarrow \varphi(x) \subseteq G$$

that is, $\varphi(U(x_0)) \subseteq G$.
(1) If $\varphi_2(x_0) \subseteq G$, there is nothing to prove. φ_2 is upper semi-continuous (u.s.c.), there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \Longrightarrow \varphi_2(x) \subseteq G$$

that is, $\varphi_2(U(x_0)) \subseteq G$ and hence $\varphi(U(x_0)) \subseteq \varphi_2(U(x_0)) \subseteq G$.
(2) Suppose that $\varphi_2(x_0) \cap (\mathbf{C}G) = K \neq \varnothing$. Let y be a point of K, $y \notin \varphi_1(x_0)$; then there exist neighbourhoods $V(y)$ and $U_y(x_0)$ such that

$$x \in U_y(x_0) \Longrightarrow \varphi_1(x) \cap V(y) = \varnothing,$$

that is, $\varphi_1(U_y(x_0)) \cap V(y) = \varnothing$.

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 上半连续对应可用“闭集上逆像为闭集”或序列判别，在紧值等假设下成立。不要把闭图像直接当成普遍等价条件。

读课件时的动作： 本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

对应的完整证明： 参见本讲义最后的「usc」证明专题。

2.48 第 48 页｜ Semi-continuous MappingsSince the set K is compact, the

Semi-continuous Mappings

Since the set K is compact, there exist elements $y_1, y_2, \cdots, y_N$ in K such that $V(y_1), V(y_2), \cdots, V(y_N)$ cover K. Write $K \subseteq \bigcup_{n=1}^{N} V(y_n) \equiv V(K)$, then $\varphi_2(x_0) \subseteq G \cup V(K)$.
If φ_2 is upper semi-continuous (u.s.c.), then there exists a neighbourhood $U'(x_0)$ such that

$$x \in U'(x_0) \Longrightarrow \varphi_2(x) \in G \cup V(K),$$

that is, $\varphi_2(U'(x_0)) \subseteq G \cup V(K)$.
Putting $U(x_0) = U_{y_1}(x_0) \cap U_{y_2}(x_0) \cap \cdots \cap U_{y_N}(x_0) \cap U'(x_0)$, we have

$$\varphi_1(U(x_0)) \cap V(y_n) \subseteq \varphi_1(U_{y_n}(x_0)) \cap V(y_n) = \varnothing,$$

that is, $\varphi_1(U(x_0)) \cap V(y_n) = \varnothing$, and hence

$$\varphi_1(U(x_0)) \cap V(K) = \varphi_1(U(x_0)) \cap \bigcup_{n=1}^{N} V(y_n) = \bigcup_{n=1}^{N} [\varphi_1(U(x_0)) \cap V(y_n)] = \varnothing$$

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：在有限维欧氏空间，紧致 $\Leftrightarrow$ 闭且有界 $\Leftrightarrow$ 每个序列有收敛子列且极限仍在集合内。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.49 第 49 页｜ Semi-continuous Mappings

Semi-continuous Mappings

Therefore

$$\begin{cases} (\varphi_1 \cap \varphi_2)(U(x_0)) \cap V(K) \subseteq \varphi_1(U(x_0)) \cap V(K) = \varnothing, \\ (\varphi_1 \cap \varphi_2)(U(x_0)) \subseteq \varphi_2(U(x_0)) \subseteq G \cup V(K), \end{cases}$$

whence

$$\varphi(U(x_0)) = (\varphi_1 \cap \varphi_2)(U(x_0)) \subseteq G,$$

that is,

$$x \in U(x_0) \Longrightarrow (\varphi_1 \cap \varphi_2)(x) \subseteq G.$$

Therefore φ is upper semi-continuous (u.s.c.). □

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

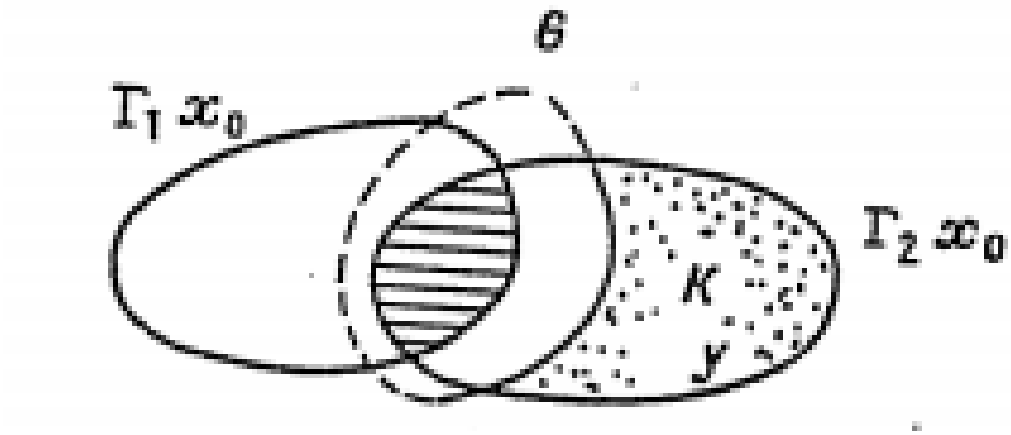
本页重点：上半连续对应可用“闭集上逆像为闭集”或序列判别，在紧值等假设下成立。不要把闭图像直接当成普遍等价条件。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论?

对应的完整证明：参见本讲义最后的「usc」证明专题。

2.50 第 50 页｜ Semi-continuous Mappings

Semi-continuous Mappings



这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

结合原文理解：读证明先辨认值集是否非空、紧、闭，以及定义域和目标空间的拓扑；缺任何一个假设都可能令结论不成立。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.51 第 51 页｜Semi-continuous Mappings

Semi-continuous Mappings

Corollary 1.1. If T is a compact space, a correspondence $\varphi : S \rightrightarrows T$ is closed-valued, then the correspondence φ is closed if and only if it is upper semi-continuous (u.s.c.).

Proof. If T is a compact space and φ is a closed-valued mapping, $\varphi(x)$ is a closed set for each $x \in S$, thus $\varphi(x)$ is a compact set and hence φ is compact-valued. If φ is u.s.c., then it is closed. Conversely, $\Delta : S \to \{T\}$ is the mapping such that $\Delta(x) = T$ for each x, Δ is a compact-valued u.s.c. correspondence. If φ is closed, then $\varphi = \varphi \cap \Delta$ is u.s.c. $\square$

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.52 第 52 页｜本页课件

Semi-continuous Mappings

Theorem 1.8. If S is a compact space and φ is a upper semi-continuous (u.s.c.) correspondence of S into itself such that, for each x, $\varphi(x) \neq \varnothing$, then there exists a non-empty compact subset K of S such that $\varphi(K) = K$.

Proof. The sets $S, \varphi(S), \varphi^2(S), \cdots$ form a sequence of non-empty compact sets; if any of these is equal to its successor, it is a set of the required kind and so the desired result is proved. We therefore assume that the sets are distinct; since $S \supseteq \varphi(S)$, we have $\varphi(S) \supseteq \varphi^2(S)$, $\varphi^2(S) \supseteq \varphi^3(S)$ and so on. Thus the sequence of sets $\varphi^n(S)$ is decreasing. By the finite intersection axiom, we have

$$K = \bigcap_{n=1}^{\infty} \varphi^n(S) \neq \varnothing.$$

For each n, we have $K \subseteq \varphi^{n-1}(S)$ and so $\varphi(K) \subseteq \varphi^n(S)$, whence $\varphi(K) \subseteq K$.

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 在有限维欧氏空间，紧致 $\Leftrightarrow$ 闭且有界 $\Leftrightarrow$ 每个序列有收敛子列且极限仍在集合内。

读课件时的动作： 本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.53 第 53 页｜Semi-continuous Mappings

Semi-continuous Mappings

We now prove that $K \subseteq \varphi(K)$. Let a be a point of K; then, for each n, there exists a point x_n in $\varphi^n(S)$ such that $a \in \varphi(x_n)$. The sequence $\{x_n : n \in \mathbb{N}\}$ admits a cluster point x_0 and a sub-sequence $\{x_{k_n} : n \in \mathbb{N}\}$ converges to x_0. Since at most $n-1$ points of this sub-sequence are outside $\varphi^n(S)$ and $\varphi^n(S)$ is compact, we have $x_0 \in \varphi^n(S)$ and so $x_0 \in K$. Also $(x_{k_n}, a) \longrightarrow (x_0, a)$ and $(x_{k_n}, a) \in G(\varphi) = \sum_{x \in S} \varphi(x)$ and so, since φ is a closed mapping, $a \in \varphi(x_0)$; therefore $a \in \varphi(K)$. It follows that $K \subseteq \varphi(K)$ and hence that $\varphi(K) = K$. $\square$

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：在有限维欧氏空间，紧致 $\Leftrightarrow$ 闭且有界 $\Leftrightarrow$ 每个序列有收敛子列且极限仍在集合内。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.54 第 54 页｜Properties of Semi-continuous Mappings

Properties of Semi-continuous Mappings

In this section we compare the properties of lower semi-continuity with those of upper semi-continuity.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

结合原文理解：读证明先辨认值集是否非空、紧、闭，以及定义域和目标空间的拓扑；缺任何一个假设都可能令结论不成立。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.55 第 55 页｜Properties of Semi-continuous Mappings

Properties of Semi-continuous Mappings

Theorem 2.1. If φ_1 is a lower semi-continuous (l.s.c.) correspondence of S into T and φ_2 is a lower semi-continuous (l.s.c.) correspondence of T into Z, the composition product $\varphi = \varphi_2 \cdot \varphi_1$ is a lower semi-continuous (l.s.c.) correspondence of S into Z.

Proof 1. If G is an open set in Z, then

$$\begin{aligned}
\varphi^-(G) &= \{x \in S \mid \varphi(x) \cap G \neq \varnothing\} \\
&= \{x \in S \mid (\varphi_2 \cdot \varphi_1)(x) \cap G \neq \varnothing\} \\
&= \{x \in S \mid \varphi_2(\varphi_1(x)) \cap G \neq \varnothing\} \\
&= \left\{x \in S \mid \varphi_1(x) \cap \varphi_2^-(G) \neq \varnothing\right\} \\
&= \varphi_1^-\left(\varphi_2^-(G)\right).
\end{aligned}$$

Therefore, if φ_1 and φ_2 are lower semi-continuous (l.s.c.), $\varphi^-(G)$ is open and so φ is lower semi-continuous (l.s.c.) □

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.56 第 56 页｜Properties of Semi-continuous Mappings

Properties of Semi-continuous Mappings

Theorem 2.1. If φ_1 is a lower semi-continuous (l.s.c.) correspondence of S into T and φ_2 is a lower semi-continuous (l.s.c.) correspondence of T into Z, then the composition product $\varphi = \varphi_2 \cdot \varphi_1$ is a lower semi-continuous (l.s.c.) correspondence of S into Z.

Proof 2. If F is a closed set in Z, then

$$\begin{aligned}
\varphi^+(F) &= \{x \in S \mid \varphi(x) \subseteq F\} \\
&= \{x \in S \mid (\varphi_2 \cdot \varphi_1)(x) \subseteq F\} \\
&= \{x \in S \mid \varphi_2(\varphi_1(x)) \subseteq F\} \\
&= \left\{x \in S \mid \varphi_1(x) \subseteq \varphi_2^+(F)\right\} \\
&= \varphi_1^+\left(\varphi_2^+(F)\right).
\end{aligned}$$

Therefore, if φ_1 and φ_2 are lower semi-continuous (l.s.c.), $\varphi^+(F)$ is closed and so φ is lower semi-continuous (l.s.c.) □

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.57 第 57 页｜Properties of Semi-continuous Mappings

Properties of Semi-continuous Mappings

Theorem 2.1′. If φ_1 is a upper semi-continuous (u.s.c.) correspondence of S into T and φ_2 is a upper semi-continuous (u.s.c.) correspondence of T into Z, then the composition product $\varphi = \varphi_2 \cdot \varphi_1$ is a upper semi-continuous (u.s.c.) correspondence of S into Z.

Proof 1. If G is an open set in Z, then

$$\begin{aligned}
\varphi^+(G) &= \{x \in S \mid \varphi(x) \subseteq G\} \\
&= \{x \in S \mid (\varphi_2 \cdot \varphi_1)(x) \subseteq G\} \\
&= \{x \in S \mid \varphi_2(\varphi_1(x)) \subseteq G\} \\
&= \left\{x \in S \mid \varphi_1(x) \subseteq \varphi_2^+(G)\right\} \\
&= \varphi_1^+\left(\varphi_2^+(G)\right).
\end{aligned}$$

Therefore, if φ_1 and φ_2 are upper semi-continuous (u.s.c.), $\varphi^+(G)$ is open and so φ is upper semi-continuous (u.s.c.) □

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.58 第 58 页｜Properties of Semi-continuous Mappings

Properties of Semi-continuous Mappings

Theorem 2.1′. If φ_1 is a upper semi-continuous (u.s.c.) correspondence of S into T and φ_2 is a upper semi-continuous (u.s.c.) correspondence of T into Z, then the composition product $\varphi = \varphi_2 \cdot \varphi_1$ is a upper semi-continuous (u.s.c.) correspondence of S into Z.

Proof 2. If F is a closed set of Z, then

$$\begin{aligned}
\varphi^{-}(F) &= \{x \in S \mid \varphi(x) \cap F \neq \varnothing\} \\
&= \{x \in S \mid (\varphi_2 \cdot \varphi_1)(x) \cap F \neq \varnothing\} \\
&= \{x \in S \mid \varphi_2(\varphi_1(x)) \cap F \neq \varnothing\} \\
&= \left\{x \in S \mid \varphi_1(x) \subseteq \varphi_2^{-}(F) \neq \varnothing\right\} \\
&= \varphi_1^{-}\left(\varphi_2^{-}(F)\right).
\end{aligned}$$

Therefore if φ_1 and φ_2 are upper semi-continuous (u.s.c.), $\varphi^{-}(F)$ is closed and so φ is upper semi-continuous (u.s.c.) □

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.59 第 59 页｜Properties of Semi-continuous Mappings

Properties of Semi-continuous Mappings

Theorem 2.2. If every mapping φ_λ of S into T is a lower semi-continuous (l.s.c.) correspondence for $\lambda \in \Lambda$, then the union $\varphi = \bigcup_{\lambda\in\Lambda} \varphi_\lambda$ is also a lower semi-continuous (l.s.c.) correspondence of S into T.

Proof 1. If G is an open set of T, we have

$$\begin{aligned}
\varphi^-(G) &= \{x \in S \mid \varphi(x) \cap G \neq \varnothing\} \\
&= \left\{x \in S \mid \bigcup_{\lambda\in\Lambda} [\varphi_\lambda(x) \cap G] \neq \varnothing\right\} \\
&= \bigcup_{\lambda\in\Lambda} \{x \in S \mid \varphi_\lambda(x) \cap G \neq \varnothing\} \\
&= \bigcup_{\lambda\in\Lambda} \varphi_\lambda^-(G).
\end{aligned}$$

Therefore the set $\varphi^-(G)$ is open and so φ is l.s.c. □

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.60 第 60 页｜本页课件

Properties of Semi-continuous Mappings

Theorem 2.2. If every mapping φ_λ of S into T is a lower semi-continuous (l.s.c.) correspondence for $\lambda \in \Lambda$, then the union $\varphi = \bigcup_{\lambda\in\Lambda} \varphi_\lambda$ is also a lower semi-continuous (l.s.c.) correspondence of S into T.

Proof 2. If F is a closed set of T, we have

$$\begin{aligned}
\varphi^+(F) &= \{x \in S \mid \varphi(x) \subseteq F\} \\
&= \left\{x \in S \mid \bigcup_{\lambda\in\Lambda} \varphi_\lambda(x) \subseteq F\right\} \\
&= \left\{x \in S \mid \mathbf{C}F \subseteq \bigcap_{\lambda\in\Lambda} \mathbf{C}(\varphi_\lambda(x))\right\} \\
&= \bigcap_{\lambda\in\Lambda} \{x \in S \mid \mathbf{C}F \subseteq \mathbf{C}(\varphi_\lambda(x))\} \\
&= \bigcap_{\lambda\in\Lambda} \{x \in S \mid \varphi_\lambda(x) \subseteq F\} \\
&= \bigcap_{\lambda\in\Lambda} \varphi_\lambda^+(F).
\end{aligned}$$

Therefore the set $\varphi^+(F)$ is closed and so φ is l.s.c. □

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作： 本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.61 第 61 页｜Properties of Semi-continuous Mappings

Properties of Semi-continuous Mappings

Theorem 2.2′. If every mapping φ_λ of S into T is a compact-valued u.s.c. correspondence for $\lambda \in \Lambda$, then the intersection $\varphi = \bigcap_{\lambda\in\Lambda} \varphi_\lambda$ is also a compact-valued u.s.c. correspondence of S into T.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：在有限维欧氏空间，紧致 $\Leftrightarrow$ 闭且有界 $\Leftrightarrow$ 每个序列有收敛子列且极限仍在集合内。

读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

2.62 第 62 页｜Properties of Semi-continuous Mappings

Properties of Semi-continuous Mappings

Proof. By Theorem 1.5, the mapping φ_λ is compact-valued u.s.c., then the mapping $\varphi = \bigcap_{\lambda\in\Lambda} \varphi_\lambda$ is closed. Suppose that $y_0 \notin \varphi(x_0) = \bigcap_{\lambda\in\Lambda} \varphi_\lambda(x_0)$, then there exists $\lambda_0 \in$ such that $y_0 \notin \varphi_{\lambda_0}(x_0)$. There exist neighbourhood $U(x_0)$ and neighbourhood $V(y_0)$ such that

$$x \in U(x_0) \qquad \Longrightarrow \qquad \varphi_{\lambda_0}(x) \cap V(y_0) = \varnothing,$$

that is, $\varphi_{\lambda_0}(U(x_0)) \cap V(y_0) = \varnothing$, then $\bigcap_{\lambda\in\Lambda} \varphi_\lambda(U(x_0)) \cap V(y_0) = \varnothing$ and hence $\varphi = \bigcap_{\lambda\in\Lambda} \varphi_\lambda$ is closed.

The intersection $\varphi = \bigcap_{\lambda\in\Lambda} \varphi_\lambda$ of a family of compact-valued mappings φ_λ is compact-valued from Non-empty Finite-intersection Theorem.

On the other hand, $\varphi = \bigcap_{\lambda\in\Lambda} \varphi_\lambda = \varphi_{\gamma'} \cap \bigcap_{\gamma\neq\gamma'} \varphi_\lambda$. The mapping $\varphi_{\gamma'}$ is compact-valued u.s.c., and the mapping $\bigcap_{\gamma\neq\gamma'} \varphi_\lambda$ is closed, therefore, by Theorem 1.7, the mapping φ is u.s.c. □

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：在有限维欧氏空间，紧致 ⇔ 闭且有界 ⇔ 每个序列有收敛子列且极限仍在集合内。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.63 第 63 页｜Properties of Semi-continuous Mappings

Properties of Semi-continuous Mappings

Theorem 2.2′. If every mapping φ_λ of S into T is a upper semi-continuous (u.s.c.) correspondence for $\lambda \in \Lambda$, then the intersection $\varphi = \bigcap_{\lambda\in\Lambda} \varphi_\lambda$ is also a upper semi-continuous (u.s.c.) correspondence of S into T.

Proof 1. If G is an open set of T, we have

$$\begin{aligned}
\varphi^+(G) &= \{x \in S \mid \varphi(x) \subseteq G\} = \left\{x \in S \mid \bigcap_{\lambda\in\Lambda} \varphi_\lambda(x) \subseteq G\right\} \\
&= \left\{x \in S \mid \mathbf{C}_T G \subseteq \mathbf{C}_T\left(\bigcap_{\lambda\in\Lambda} \varphi_\lambda(x)\right)\right\} = \left\{x \in S \mid \mathbf{C}_T G \subseteq \bigcup_{\lambda\in\Lambda} \mathbf{C}_T\left(\varphi_\lambda(x)\right)\right\} \\
&= \bigcup_{\lambda\in\Lambda} \{x \in S \mid \mathbf{C}_T G \subseteq \mathbf{C}_T\left(\varphi_\lambda(x)\right)\} = \bigcup_{\lambda\in\Lambda} \{x \in S \mid \varphi_\lambda(x) \subseteq G\} = \bigcup_{\lambda\in\Lambda} \varphi_\lambda^+(G)
\end{aligned}$$

Therefore the set $\varphi^+(G)$ is open and so φ is u.s.c. □

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.64 第 64 页｜本页课件

Properties of Semi-continuous Mappings

Theorem 2.2′. If every mapping φ_λ of S into T is a upper semi-continuous (u.s.c.) correspondence for $\lambda \in \Lambda$, then the intersection $\varphi = \bigcap_{\lambda\in\Lambda} \varphi_\lambda$ is also a upper semi-continuous (u.s.c.) correspondence of S into T.

Proof 2. If F is a closed set of T, we have

$$
\begin{aligned}
\varphi^-(F) &= \{x \in S \mid \varphi(x) \cap F \neq \varnothing\} = \left\{x \in S \mid \bigcap_{\lambda\in\Lambda} \varphi_\lambda(x) \cap F \neq \varnothing\right\} \\
&= \mathbf{C}_S \left\{x \in S \mid \bigcap_{\lambda\in\Lambda} \varphi_\lambda(x) \cap F = \varnothing\right\} = \mathbf{C}_S \left\{x \in S \mid F \subseteq \mathbf{C}_T \left(\bigcap_{\lambda\in\Lambda} \varphi_\lambda(x)\right)\right\} \\
&= \mathbf{C}_S \left\{x \in S \mid F \subseteq \bigcup_{\lambda\in\Lambda} \mathbf{C}_T\left(\varphi_\lambda(x)\right)\right\} = \mathbf{C}_S \left(\bigcup_{\lambda\in\Lambda} \{x \in S \mid F \subseteq \mathbf{C}_T\left(\varphi_\lambda(x)\right)\}\right) \\
&= \mathbf{C}_S \left(\bigcup_{\lambda\in\Lambda} \{x \in S \mid \varphi_\lambda(x) \subseteq \mathbf{C}_T F\}\right) = \mathbf{C}_S \left(\bigcup_{\lambda\in\Lambda} \{x \in S \mid \varphi_\lambda(x) \cap F = \varnothing\}\right) \\
&= \bigcap_{\lambda\in\Lambda} \{x \in S \mid \varphi_\lambda(x) \cap F \neq \varnothing\} = \bigcap_{\lambda\in\Lambda} \varphi_\lambda^-(F).
\end{aligned}
$$

Therefore the set $\varphi^-(F)$ is closed and so φ is u.s.c. □

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作： 本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.65 第 65 页｜ Properties of Semi-continuous MappingsPseudo-Theorem 2.

Properties of Semi-continuous Mappings

Pseudo-Theorem 2.3. If every mapping φ_n of S into T is a lower semi-continuous (l.s.c.) correspondence for $n = 1, \cdots, N$, then the intersection $\varphi = \bigcap_{n=1}^{N} \varphi_n$ is also a lower semi-continuous (l.s.c.) correspondence of S into T.

Proof 1. If G is an open set in T, we have

$$
\begin{aligned}
\varphi^{-}(G) &= \{x \in S \mid \varphi(x) \cap G \neq \varnothing\} = \left\{x \in S \mid \bigcap_{n=1}^{N} \varphi_n(x) \cap G \neq \varnothing\right\} \\
&= \mathbf{C}_S \left\{x \in S \mid \bigcap_{n=1}^{N} \varphi_n(x) \cap G = \varnothing\right\} = \mathbf{C}_S \left\{x \in S \mid \bigcap_{n=1}^{N} \varphi_n(x) \subseteq \mathbf{C}_T G\right\} \\
&= \mathbf{C}_S \left\{x \in S \mid G \subseteq \bigcup_{n=1}^{N} \mathbf{C}_T\left(\varphi_n(x)\right)\right\} = \mathbf{C}_S \left(\bigcup_{n=1}^{N} \{x \in S \mid G \subseteq \mathbf{C}_T\left(\varphi_n(x)\right)\}\right) \\
&= \mathbf{C}_S \left(\bigcup_{n=1}^{N} \{x \in S \mid \varphi_n(x) \subseteq \mathbf{C}_T G\}\right) = \mathbf{C}_S \left(\bigcup_{n=1}^{N} \{x \in S \mid \varphi_n(x) \cap G = \varnothing\}\right) \\
&= \bigcap_{n=1}^{N} \{x \in S \mid \varphi_n(x) \cap G \neq \varnothing\} = \bigcap_{n=1}^{N} \varphi_n^{-}(G).
\end{aligned}
$$

Therefore $\varphi^{-}(G)$ is an open set and so φ is l.s.c. □

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作： 本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.66 第 66 页｜ Properties of Semi-continuous MappingsPseudo-Theorem 2.

Properties of Semi-continuous Mappings

Pseudo-Theorem 2.3. If every mapping φ_n of S into T is a lower semi-continuous (l.s.c.) correspondence for $n = 1, \cdots, N$, then the intersection $\varphi = \bigcap_{n=1}^{N} \varphi_n$ is also a lower semi-continuous (l.s.c.) correspondence of S into T.

Proof 2. If F is a closed set in T, we have

$$\begin{aligned}
\varphi^+(F) &= \{x \in S \mid \varphi(x) \subseteq F\} = \left\{x \in S \mid \bigcap_{n=1}^{N} \varphi_n(x) \subseteq F\right\} \\
&= \left\{x \in S \mid \mathbf{C}_T F \subseteq \mathbf{C}_T \left(\bigcap_{n=1}^{N} \varphi_n(x)\right)\right\} \\
&= \left\{x \in S \mid \mathbf{C}_T F \subseteq \bigcup_{n=1}^{N} \mathbf{C}_T\left(\varphi_n(x)\right)\right\} \\
&= \bigcup_{n=1}^{N} \{x \in S \mid \mathbf{C}_T F \subseteq \mathbf{C}_T\left(\varphi_n(x)\right)\} \\
&= \bigcup_{n=1}^{N} \{x \in S \mid \varphi_n(x) \subseteq F\} = \bigcup_{n=1}^{N} \varphi_n^+(F).
\end{aligned}$$

Therefore $\varphi^+(F)$ is a closed set and so φ is l.s.c. □

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作： 本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.67 第 67 页｜Properties of Semi-continuous Mappings

Properties of Semi-continuous Mappings

Theorem 2.3′. If every mapping φ_n of S into T is a compacted-valued u.s.c. correspondence for $n = 1, \cdots, N$, then the union $\varphi = \bigcup_{n=1}^{N} \varphi_n$ is also a compacted-valued u.s.c. correspondence of S into T.

Proof 1. The set $\varphi(x) = \bigcup_{n=1}^{N} \varphi_n(x)$ is compact for all x. Moreover, if G is an open set in T, we have

$$\begin{aligned}
\varphi^+(G) &= \{x \in S \mid \varphi(x) \subseteq G\} = \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \subseteq G\right\} \\
&= \left\{x \in S \mid \mathbf{C}_T G \subseteq \mathbf{C}_T \left(\bigcup_{n=1}^{N} \varphi_n(x)\right)\right\} = \left\{x \in S \mid \mathbf{C}_T G \subseteq \bigcap_{n=1}^{N} \mathbf{C}_T\left(\varphi_n(x)\right)\right\} \\
&= \bigcap_{n=1}^{N} \{x \in S \mid \mathbf{C}_T G \subseteq \mathbf{C}_T\left(\varphi_n(x)\right)\} = \bigcap_{n=1}^{N} \{x \in S \mid \varphi_n(x) \subseteq G\} = \bigcap_{n=1}^{N} \varphi_n^+(G).
\end{aligned}$$

Therefore $\varphi^+(G)$ is an open set and so φ is u.s.c. □

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.68 第 68 页｜Properties of Semi-continuous Mappings

Properties of Semi-continuous Mappings

Theorem 2.3′. If every mapping φ_n of S into T is a compacted-valued u.s.c. correspondence for $n = 1, \cdots, N$, then the union $\varphi = \bigcup_{n=1}^{N} \varphi_n$ is also a compacted-valued u.s.c. correspondence of S into T.

Proof 2. The set $\varphi(x) = \bigcup_{n=1}^{N} \varphi_n(x)$ is compact for all x. Moreover, if F is a closed set in T, we have

$$\begin{aligned}
\varphi^{-}(F) &= \{x \in S \mid \varphi(x) \cap F \neq \varnothing\} = \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \cap F \neq \varnothing\right\} \\
&= \mathbf{C}_S \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \cap F = \varnothing\right\} = \mathbf{C}_S \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \subseteq \mathbf{C}_T F\right\} \\
&= \mathbf{C}_S \left(\bigcup_{n=1}^{N} \{x \in S \mid \varphi_n(x) \subseteq \mathbf{C}_T F\}\right) = \mathbf{C}_S \left(\bigcup_{n=1}^{N} \{x \in S \mid \varphi_n(x) \cap F \neq \varnothing\}\right) \\
&= \bigcap_{n=1}^{N} \{x \in S \mid \varphi_n(x) \cap F = \varnothing\} = \bigcap_{n=1}^{N} \varphi_n^{-}(F).
\end{aligned}$$

Therefore $\varphi^{-}(F)$ is a closed set and so φ is u.s.c. □

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.69 第 69 页｜ Properties of Semi-continuous MappingsTheorem 2.3′. If

Properties of Semi-continuous Mappings

Theorem 2.3′. If every mapping φ_n of S into T is a upper semi-continuous (u.s.c.) correspondence for $n = 1, \cdots, N$, then the union $\varphi = \bigcup_{n=1}^{N} \varphi_n$ is also a upper semi-continuous (u.s.c.) correspondence of S into T.

Proof 1. If G is an open set in T, we have

$$\begin{aligned}
\varphi^+(G) &= \{x \in S \mid \varphi(x) \subseteq G\} = \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \subseteq G\right\} \\
&= \left\{x \in S \mid \mathbf{C}_T G \subseteq \mathbf{C}_T \left(\bigcup_{n=1}^{N} \varphi_n(x)\right)\right\} \\
&= \left\{x \in S \mid \mathbf{C}_T G \subseteq \bigcap_{n=1}^{N} \mathbf{C}_T\left(\varphi_n(x)\right)\right\} \\
&= \bigcap_{n=1}^{N} \{x \in S \mid \mathbf{C}_T G \subseteq \mathbf{C}_T\left(\varphi_n(x)\right)\} \\
&= \bigcap_{n=1}^{N} \{x \in S \mid \varphi_n(x) \subseteq G\} = \bigcap_{n=1}^{N} \varphi_n^+(G).
\end{aligned}$$

Therefore $\varphi^+(G)$ is an open set and so φ is u.s.c. □

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作： 本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.70 第 70 页｜Properties of Semi-continuous Mappings

Properties of Semi-continuous Mappings

Theorem 2.3′. If every mapping φ_n of S into T is a upper semi-continuous (u.s.c.) correspondence for $n = 1, \cdots, N$, then the union $\varphi = \bigcup_{n=1}^{N} \varphi_n$ is also a upper semi-continuous (u.s.c.) correspondence of S into T.

Proof 2. If F is a closed set in T, we have

$$\begin{aligned}
\varphi^-(F) &= \{x \in S \mid \varphi(x) \cap F \neq \varnothing\} = \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \cap F \neq \varnothing\right\} \\
&= \mathbf{C}_S \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \cap F = \varnothing\right\} = \mathbf{C}_S \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \subseteq \mathbf{C}_T F\right\} \\
&= \mathbf{C}_S \left(\bigcup_{n=1}^{N} \{x \in S \mid \varphi_n(x) \subseteq \mathbf{C}_T F\}\right) = \mathbf{C}_S \left(\bigcup_{n=1}^{N} \{x \in S \mid \varphi_n(x) \cap F \neq \varnothing\}\right) \\
&= \bigcap_{n=1}^{N} \{x \in S \mid \varphi_n(x) \cap F = \varnothing\} = \bigcap_{n=1}^{N} \varphi_n^-(F).
\end{aligned}$$

Therefore $\varphi^-(F)$ is a closed set and so φ is u.s.c. □

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.71 第 71 页｜本页课件

Properties of Semi-continuous Mappings

Theorem 2.4. If every mapping φ_n is a lower semi-continuous (l.s.c.) correspondence of S into T_n for $n = 1, \cdots, N$, then the Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n$ is also a lower semi-continuous (l.s.c.) correspondence of S into $T = \prod_{n=1}^{N} T_n$.

Proof 1. Let $G = \bigcup_k E^k$ be the union of the elementary open sets E^k in T. Since $E^k = \prod_{n=1}^{N} G_n^k$, where G_n^k is an open set in T_n, we have

$$\begin{aligned}
\varphi^- \left(E^k\right) &= \left\{x \in S \mid \varphi(x) \cap E^k \neq \varnothing\right\} = \left\{x \in S \mid \prod_{n=1}^{N} \varphi_n(x) \cap \prod_{n=1}^{N} G_n^k \neq \varnothing\right\} \\
&= \left\{x \in S \mid \prod_{n=1}^{N} \left(\varphi_n(x) \cap G_n^k\right) \neq \varnothing\right\} = \bigcap_{n=1}^{N} \left\{x \in S \mid \varphi_n(x) \cap G_n^k \neq \varnothing\right\} = \bigcap_{n=1}^{N} \varphi_n^- \left(G_n^k\right).
\end{aligned}$$

Then $\varphi^- \left(E^k\right)$ is an open set and therefore so is $\varphi^-(G) = \bigcup_k \varphi^- \left(E^k\right)$.

$$\begin{aligned}
\varphi^-(G) &= \varphi^- \left(\bigcup_k E^k\right) = \left\{x \in S \mid \varphi(x) \cap \left(\bigcup_k E^k\right) \neq \varnothing\right\} \\
&= \left\{x \in S \mid \bigcup_k \left[\varphi(x) \cap E^k\right] \neq \varnothing\right\} \\
&= \bigcup_k \left\{x \in S \mid \varphi(x) \cap E^k \neq \varnothing\right\} = \bigcup_k \varphi^- \left(E^k\right).
\end{aligned}$$

Hence φ is l.s.c. □

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作： 本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.72 第 72 页｜本页课件

Properties of Semi-continuous Mappings

Theorem 2.4. If every mapping φ_n is a lower semi-continuous (l.s.c.) correspondence of S into T_n for $n = 1, \cdots, N$, then the Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n$ is also a lower semi-continuous (l.s.c.) correspondence of S into $T = \prod_{n=1}^{N} T_n$.

Proof 2. Let $F = \bigcap_k E^k$ be the intersection of the elementary closed sets E^k in T. Since $E^k = \prod_{n=1}^{N} F_n^k$, where F_n^k is a closed set in T_n, we have

$$\begin{aligned} \varphi^+\left(E^k\right) &= \left\{x \in S \mid \varphi(x) \subseteq E^k\right\} = \left\{x \in S \mid \prod_{n=1}^{N} \varphi_n(x) \subseteq \prod_{n=1}^{N} F_n^k\right\} \\ &= \bigcap_{n=1}^{N} \left\{x \in S \mid \varphi_n(x) \subseteq F_n^k\right\} = \bigcap_{n=1}^{N} \varphi_n^+\left(F_n^k\right). \end{aligned}$$

Then $\varphi^+\left(E^k\right)$ is a closed set and therefore so is $\varphi^+(F) = \bigcap_k \varphi^+\left(E^k\right)$.

$$\begin{aligned} \varphi^+(F) &= \varphi^+\left(\bigcap_k E^k\right) = \left\{x \in S \mid \varphi(x) \subseteq \bigcap_k E^k\right\} \\ &= \bigcap_k \left\{x \in S \mid \varphi(x) \subseteq E^k\right\} = \bigcap_k \varphi^+\left(E^k\right). \end{aligned}$$

Hence φ is l.s.c. □

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作： 本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.73 第 73 页｜本页课件

Properties of Semi-continuous Mappings

Theorem 2.4′. If every mapping φ_n is a upper semi-continuous (u.s.c.) correspondence of S into T_n for $n = 1, \cdots, N$, then the Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n$ is also a upper semi-continuous (u.s.c.) correspondence of S into $T = \prod_{n=1}^{N} Y_n$.

Proof 1. Let $G = \bigcup_k E^k$ be the union of the elementary open sets E^k in T. Since $E^k = \prod_{n=1}^{N} G_n^k$, where G_n^k is an open set in T_n, we have

$$\begin{aligned} \varphi^+\left(E^k\right) &= \left\{x \in S \mid \varphi(x) \subseteq E^k\right\} = \left\{x \in S \mid \prod_{n=1}^{N} \varphi_n(x) \subseteq \prod_{n=1}^{N} G_n^k\right\} \\ &= \bigcap_{n=1}^{N} \left\{x \in S \mid \varphi_n(x) \subseteq G_n^k\right\} = \bigcap_{n=1}^{N} \varphi_n^+\left(G_n^k\right). \end{aligned}$$

Then $\varphi^+\left(E^k\right)$ is an open set and therefore so is $\varphi^+(G) = \bigcup_k \varphi^+\left(E^k\right)$.

$$\begin{aligned} \varphi^+(G) &= \varphi^+\left(\bigcup_k E^k\right) = \left\{x \in S \mid \varphi(x) \subseteq \bigcup_k E^k\right\} \\ &= \bigcup_k \left\{x \in S \mid \varphi(x) \subseteq E^k\right\} = \bigcup_k \varphi^+\left(E^k\right). \end{aligned}$$

Hence φ is u.s.c. □

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作： 本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.74 第 74 页｜本页课件

Properties of Semi-continuous Mappings

Theorem 2.4′. If every mapping φ_n is a upper semi-continuous (u.s.c.) correspondence of S into T_n for $n = 1, \cdots, N$, then the Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n$ is also a upper semi-continuous (u.s.c.) correspondence of S into $T = \prod_{n=1}^{N} T_n$.

Proof 2. Let $F = \bigcap_k E^k$ be the intersection of the elementary closed sets E^k in T. Since $E^k = \prod_{n=1}^{N} F_n^k$, where F_n^k is a closed set in T_n, we have

$$\begin{aligned}\varphi^-\left(E^k\right) &= \left\{x \in S \mid \varphi(x) \cap E^k \neq \varnothing\right\} = \left\{x \in S \mid \prod_{n=1}^{N} \varphi_n(x) \cap \prod_{n=1}^{N} F_n^k \neq \varnothing\right\} \\ &= \left\{x \in S \mid \prod_{n=1}^{N} \left(\varphi_n(x) \cap F_n^k\right) \neq \varnothing\right\} = \bigcap_{n=1}^{N} \left\{x \in S \mid \varphi_n(x) \cap F_n^k \neq \varnothing\right\} = \bigcap_{n=1}^{N} \varphi_n^-\left(F_n^k\right).\end{aligned}$$

Then $\varphi^-\left(E^k\right)$ is a closed set and therefore so is $\varphi^-(F) = \bigcap_k \varphi^-\left(E^k\right)$.

$$\begin{aligned}\varphi^-(F) &= \varphi^-\left(\bigcap_k E^k\right) = \left\{x \in S \mid \varphi(x) \cap \left(\bigcap_k E^k\right) \neq \varnothing\right\} \\ &= \left\{x \in S \mid \bigcap_k \left[\varphi(x) \cap E^k\right] \neq \varnothing\right\} \\ &= \bigcap_k \left\{x \in S \mid \varphi(x) \cap E^k \neq \varnothing\right\} = \bigcap_k \varphi^-\left(E^k\right).\end{aligned}$$

Hence φ is u.s.c. □

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作： 本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.75 第 75 页｜Properties of Semi-continuous Mappings

Properties of Semi-continuous Mappings

Theorem 2.4′. If every mapping φ_n is a compact-valued u.s.c. correspondence of S into T_n for $n = 1, \cdots, N$, then the Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n$ is also a compact-valued u.s.c. correspondence of S into $T = \prod_{n=1}^{N} T_n$.

Proof. To fix our ideas, we consider the case $N = 2$. Since $\varphi_1(x)$ and $\varphi_2(x)$ are compact, so is $\varphi(x) = \varphi_1(x) \times \varphi_2(x)$, by Tychonoff's Theorem.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：在有限维欧氏空间，紧致 $\Leftrightarrow$ 闭且有界 $\Leftrightarrow$ 每个序列有收敛子列且极限仍在集合内。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.76 第 76 页｜本页课件

Properties of Semi-continuous Mappings

Let G be an open set in T and. let a be an element of $\varphi^+(G)$. Since $\varphi(a)$ is compact and is contained in G, it can be covered by a finite number of elementary open sets contained in G; let $E^1, \cdots, E^K$ be such a finite covering. Suppose that $(y_1, y_2) \in \varphi_1(a) \times \varphi_2(a)$; let $E(y_1)$ denote the union of the E^k meeting the set $\{y_1\} \times \varphi_2(a)$ and let $E(y_2)$ be the union of the E^k meeting the set $\varphi_1(a) \times \{y_2\}$. The projection $\pi_2 E(y_1)$ of the set $E(y_1)$ on the space Y_2 is open, by Theorem 2 of §9, Chapter IV. As y_1 varies the sets. $\pi_2 E(y_1)$ are finite in number and therefore their intersection is open; similarly the intersection of the sets $\pi_1 E(y_2)$ is open. We write

$$G_1 = \bigcap_{y_2 \in \varphi_2(a)} \pi_1 E(y_2) \qquad \text{and} \qquad G_2 = \bigcap_{y_1 \in \varphi_1(a)} \pi_2 E(y_1).$$

Then $E = G_1 \times G_2$ is an elementary open set of T and

$$\varphi^+(E) = \{x \mid \varphi_1(x) \times \varphi_2(x) \subseteq G_1 \times G_2\} = \left[\varphi_1^+(G_1) \cap \varphi_2^+(G_2)\right] \cup \varphi_1^+(\varnothing) \cup \varphi_2^+(\varnothing).$$

Therefore $\varphi^+(E)$ is an open set of S; since $\varphi(a) \subseteq E \subseteq G$, we also have

$$a \in \varphi^+(E) \subseteq \varphi^+(G)$$

Therefore the set $\varphi^+(G)$ is a neighbourhood of each of its points and so is open; hence φ is compact-valued u.s.c. □

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作： 本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

2.77 第 77 页｜Properties of Semi-continuous Mappings

Properties of Semi-continuous Mappings

Theorem 2.5′. Let S and T be topological vector spaces. If every mapping $\varphi_n : X \to Y$ is a compact-valued u.s.c. correspondence for $n = 1, \cdots, N$, then the sum $\varphi = \sum_{n=1}^{N} \varphi_n$ is a compact-valued u.s.c. correspondence.

Theorem 2.5′. The sum $\varphi = \sum_{n=1}^{N} \varphi_n$ of a finite number of compact-valued u.s.c. correspondences φ_n of S into T is a compact-valued u.s.c. correspondence of S into T.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：在有限维欧氏空间，紧致 $\Leftrightarrow$ 闭且有界 $\Leftrightarrow$ 每个序列有收敛子列且极限仍在集合内。

读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

2.78 第 78 页｜Properties of Semi-continuous Mappings

Properties of Semi-continuous Mappings

Proof. We use mathematical induction to verify this theorem.

(1) If $\varphi_n(x_0)$ is compact for $n = 1, \cdots, N$, then $\varphi(x_0) = \sum_{n=1}^{N} \varphi_n(x_0)$ is compact.

(2) For any open set $G \subseteq T$ with $\varphi(x_0) = \sum_{n=1}^{N} \varphi_n(x_0) \subseteq G$, there exists a neighborhood $U(x_0)$ such that $x \in U(x_0)$ implies $\varphi(x) = \sum_{n=1}^{N} \varphi_n(x) \subseteq G$.

(1) Let T be a topological vector space. Define $f : T^N \to T$ by $f(y_1, \cdots, y_N) = \sum_{n=1}^{N} y_n$, then f is a continuous function, $f(\varphi_1 x_0, \cdots, \varphi_n(x_0)) = \sum_{n=1}^{N} \varphi_n(x_0)$. As you know, $\varphi_n(x_0)$ is compact for $n = 1, \cdots, N$, then $\varphi(x_0) = \sum_{n=1}^{N} \varphi_n(x_0)$ is compact.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.79 第 79 页｜Properties of Semi-continuous Mappings

Properties of Semi-continuous Mappings

(2) For any open set $G \subseteq T$ with $\varphi(x_0) = \sum_{n=1}^{N} \varphi_n(x_0) \subseteq G$, there exists a neighborhood $V = U(0)$ of origin such that $\sum_{n=1}^{N} \varphi_n(x_0) + V \subseteq G$, there exist N neighborhoods $V_1, \cdots, V_N$ with $\sum_{n=1}^{N} V_n \subseteq V$ such that
$\sum_{n=1}^{N} [\varphi_n(x_0) + V_n] = \sum_{n=1}^{N} \varphi_n(x_0) + V \subseteq G$. $\varphi_n : X \to Y$ is u.s.c. for $n = 1, \cdots, N$, then $\varphi_n(x_0) \subseteq \varphi_n(x_0) + V_n$, there exists a neighborhood $U_n(x_0)$ such that

$$x \in U_n(x_0) \quad \Longrightarrow \quad x \in \varphi_n(x_0) + V_n.$$

For any $x \in N(x_0) \equiv \bigcap_{n=1}^{N} U_n(x_0)$,

$$\varphi(x) = \sum_{n=1}^{N} \varphi_n(x) \subseteq \sum_{n=1}^{N} [\varphi_n(x_0) + V_n] \subseteq \sum_{n=1}^{N} \varphi_n(x_0) + V \subseteq G.$$

Thus $\varphi = \sum_{n=1}^{N} \varphi_n$ is compact-valued and u.s.c. □

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.80 第 80 页｜Properties of Semi-continuous Mappings

Properties of Semi-continuous Mappings

Theorem 2.6. Let T be a linear vector space. If $\varphi : S \to \mathbb{R}^N$ is a lower semi-continuous (l.s.c.) correspondence, then its convex envelope $\mathbf{CE}(\varphi)$ is a lower semi-continuous (l.s.c.) correspondence.

Proof. According to Caratheodory Characterization Theorem, for any $y_0 \in \mathbf{CE}(\varphi)(x_0) = \mathbf{CE}(\varphi(x_0))$,

$$y_0 = \sum_{k=1}^{K} \lambda^k z_0^k, \qquad z_0^k \in \varphi(x_0), \quad \sum_{k=1}^{K} \lambda_n^k = 1, \quad \lambda_n^k \geqslant 0.$$

For any $x_n \to x_0$, there exists a sequence $z_n^k \to z_0^k$ such that $z_n^k \in \varphi(x_n)$, $y_n = \sum_{k=1}^{K} \lambda^k z_n^k \in \varphi(x_0) \subseteq \mathbf{CE}(\varphi(x_n))$. Therefore $\mathbf{CE}(\varphi)$ is lower semi-continuous (l.s.c.). □

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：下半连续要求：若 y Φ(x)，当自变量序列趋近给定点时，可以从临近的对应值集中挑选一列点逼近原来的目标点（在适当的度量空间设定下）。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

对应的完整证明：参见本讲义最后的「lsc」证明专题。

2.81 第 81 页｜Properties of Semi-continuous Mappings

Properties of Semi-continuous Mappings

Theorem 2.6′. If $\varphi : S \to \mathbb{R}^N$ is a compact-valued u.s.c. correspondence, then its convex envelope $\mathbf{CE}(\varphi)$ is a compact-valued u.s.c. correspondence.

Proof. According to Caratheodory Characterization Theorem, for any $y_n \in \mathbf{CE}(\varphi)(x_n) = \mathbf{CE}(\varphi(x_n))$,

$$y_n = \sum_{k=1}^{K} \lambda_n^k z_n^k, \qquad z_n^k \in \varphi(x_n), \quad \sum_{k=1}^{K} \lambda_n^k = 1, \quad \lambda_n^k \geqslant 0.$$

For any $x_n \to x_0$, there exists a convergent subsequence $\{z_{n'}^k, n' \in \mathbb{N}\} \subseteq \{z_n^k, n \in \mathbb{N}\}$ such that $\lim_{n'} z_{n'}^k \to z_0^k \in \varphi(x_0)$, then there corresponds a subsequence $\{y_{n'}, n' \in \mathbb{N}\} \subseteq \{y_n, n \in \mathbb{N}\}$ such that $y_{n'} = \sum_{k=1}^{K} \lambda_{n'}^k z_{n'}^k \to \sum_{k=1}^{K} \lambda_n^k z_0^k = y_0 \in \varphi(x_0) \subseteq \mathbf{CE}(\varphi(x_0))$. Therefore $\mathbf{CE}(\varphi)$ is compact-valued and u.s.c. □

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 在有限维欧氏空间，紧致 ⇔ 闭且有界 ⇔ 每个序列有收敛子列且极限仍在集合内。

读课件时的动作： 本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.82 第 82 页｜Properties of Semi-continuous Mappings

Properties of Semi-continuous Mappings

Theorem 2.7. $\varphi : S \to T$ is a lower semi-continuous (l.s.c.) correspondence if and only if its convex hull $\mathbf{CH}(\varphi) = \overline{\varphi}$ is a lower semi-continuous (l.s.c.) correspondence.

Proof. For any open set $G \subseteq T$ with $\varphi(x_0) \subseteq G$ equivalently $\overline{\varphi}(x_0) \subseteq G$, there exists a neighbourhood $U(x_0)$ such that, for any $x \in U(x_0)$, $\varphi(x) \subseteq G$ equivalently $\overline{\varphi}(x) \subseteq G$. Thus φ is l.s.c. if and only if its convex hull $\mathbf{CH}(\varphi) = \overline{\varphi}$ is l.s.c. □

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.83 第 83 页｜Properties of Semi-continuous Mappings

Properties of Semi-continuous Mappings

Theorem 2.7′. Let T be a regular metric space. If $\varphi : S \to T$ is a u.s.c. correspondence, then its convex hull $\mathbf{CH}(\varphi) = \overline{\varphi}$ is a u.s.c. correspondence.

Proof. For any open set $G \subseteq T$ with $\overline{\varphi}(x_0) \subseteq G$, $\overline{\varphi}(x_0) \cap (\mathbf{C}_T G) = \varnothing$. According to the regularity of metric space, there exists an open set G' and O such that

$$\overline{\varphi}(x_0) \subseteq G', \qquad \mathbf{C}_T G \subseteq O, \qquad G' \cap O = \varnothing$$

As φ is u.s.c., there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \subseteq G',$$

that is, $\varphi(U(x_0)) \subseteq G'$. Since $G' \cap (\mathbf{C}_T G) = \varnothing$, then $\overline{\varphi}(U(x_0)) \subseteq G$. Therefore $\overline{\varphi}$ is u.s.c. □

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.84 第 84 页｜Semi-continuous Correspondences

Semi-continuous Correspondences

Definition 3.1 Let X and Y be metric spaces, and let $\varphi : X \rightrightarrows Y$ be a correspondence.

(1) We say that φ is inner semi-continuous (i.s.c.) if whenever $x_n \to x$ and $y \in \varphi(x)$, there is a sequence $y_n \in \varphi(x_n)$ for each n such that $y_n \to y$.

(2) We say that φ is outer semi-continuous (o.s.c.) if whenever $x_n \to x$ and $y_n \in \varphi(x_n)$ for each n, there is a convergent subsequence $y_{n_k} \to y$ with $y \in \varphi(x)$.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

结合原文理解：读证明先辨认值集是否非空、紧、闭，以及定义域和目标空间的拓扑；缺任何一个假设都可能令结论不成立。

读课件时的动作：本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.85 第 85 页｜Semi-continuous Correspondences

Semi-continuous Correspondences

Theorem 3.1 (Lower Semi-continuity and Inner Semi-continuity) Let X and Y be two metric spaces. A correspondence $\varphi : X \rightrightarrows Y$. φ is l.s.c. at x if and only if it is i.s.c. at x.

Proof. Suppose φ is not l.s.c. at some $x_0 \in X$. Then there exists an open subset G of Y such that $\varphi(x_0) \cap G \neq \varnothing$ and, for every $n \in \mathbb{N}$, there exists an $x_n \in B_{\frac{1}{n}}(x_0)$ with $\varphi(x_n) \cap G = \varnothing$. Note that $x_n \to x_0$, and pick any $y_0 \in \varphi(x_0) \cap G$. By hypothesis, there must exist a sequence $\{y_n : n \in \mathbb{N}\} \subseteq Y$ such that $y_n \to y_0$ and $y_n \in \varphi(x_n)$ for each n. But since $y_0 \in G$ and G is open, $y_n \in G$ for n large enough, contradicting that $\varphi(x_n) \cap G = \varnothing$ for all n.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：下半连续要求：若 y Φ(x)，当自变量序列趋近给定点时，可以从临近的对应值集中挑选一列点逼近原来的目标点（在适当的度量空间设定下）。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

对应的完整证明：参见本讲义最后的「lsc」证明专题。

2.86 第 86 页｜Semi-continuous Correspondences

Semi-continuous Correspondences

Conversely, if $x_n \to x_0$ and $y_0 \in \varphi(x_0)$, then $y_0 \in \varphi(x_0) \cap B_{\frac{1}{n}}(y_0)$. By lower semi-continuity, there exists a $\varepsilon_n > 0$ with $\lim_{n\to\infty} \varepsilon_n = 0$ such that

$$x \in B_{\varepsilon_n}(x_0) \quad \Longrightarrow \quad \varphi(x) \cap B_{\frac{1}{n}}(y_0) \neq \varnothing.$$

Since $x_n \to x_0$, so there exists k_n such that for all $n \geqslant k_n$, $x_n \in B_{\varepsilon_n}(x_0)$, then $\varphi(x_n) \cap B_{\frac{1}{n}}(y_0) \neq \varnothing$. We may assume k_n is increasing.

For $n < k_n$, we pick $y_n \in \varphi(x_n)$. For any $k_n \leqslant n < k_{n+1}$, we have $\varphi(x_n) \cap B_{\frac{1}{n}}(y_0) \neq \varnothing$, pick $y_n \in \varphi(x_n) \cap B_{\frac{1}{n}}(y_0)$. Then $d(y_n, y_0) < \dfrac{1}{n}$. $y_n \to y_0$. Hence φ is i.s.c. at x_0. $\square$

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

结合原文理解：读证明先辨认值集是否非空、紧、闭，以及定义域和目标空间的拓扑；缺任何一个假设都可能令结论不成立。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.87 第 87 页｜Semi-continuous Correspondences

Semi-continuous Correspondences

Theorem 3.1′ (Upper Semi-continuity and outer Semi-continuity) Let X and Y be two metric spaces. Mapping $\varphi : X \rightrightarrows Y$ is a correspondence.

(1) If φ is compact-valued u.s.c. at x, then it is o.s.c. at x.

(2) If φ is o.s.c. at x, then φ is u.s.c. at x.

Proof of (1). Suppose φ is u.s.c. at x_0, $x_n \to x_0$ and $y_n \in \varphi(x_n)$. Since φ is compact-valued, $\varphi(x_0)$ has a bounded neighborhood U. Then $S = \{x_0, x_1, x_2, \cdots\}$ is sequentially compact in X, hence S is compact in X. By Proposition 1, $\varphi(S)$ is sequentially compact in Y. So, if $y_n \in \varphi(x_n)$ for each n, then $\{y_n : n \in \mathbb{N}\}$ — being a sequence in $\varphi(S)$ – must possess a subsequence that converges to some $y_0 \in \varphi(S)$. With an innocent abuse of notation, we denote this subsequence also by $\{y_n : n \in \mathbb{N}\}$, and write $y_n \to y_0$.

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 在有限维欧氏空间，紧致 $\Leftrightarrow$ 闭且有界 $\Leftrightarrow$ 每个序列有收敛子列且极限仍在集合内。

读课件时的动作： 本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.88 第 88 页｜Semi-continuous Correspondences

Semi-continuous Correspondences

We wish to show that $y_0 \in \varphi(x_0)$. Let $y_0 \notin \varphi(x_0)$, and observe that this implies $\varepsilon = d_Y(y_0, \varphi(x_0)) > 0$ because $\varphi(x_0)$ is a compact (hence closed) set. Now define

$$T = \left\{ z \in Y : d_Y(z, \varphi(x_0)) \leqslant \frac{\varepsilon}{2} \right\}$$

Since $d_Y(z, \varphi(x_0))$ is a continuous map on z, T is a closed set. Moreover, by definition, we have

$$\varphi(x_0) \subseteq T^\circ$$

and $y_0 \notin T$. But, since φ is u.s.c. at x_0, there exists $\delta > 0$ such that

$$\varphi(B_\delta(x_0)) \subseteq T^\circ.$$

Thus there is $n \in \mathbb{N}$ such that $y_n \in \varphi(x_n) \subseteq T^\circ$ for all $n \geqslant N$. But then, since T is a closed set and $y_n \to y_0$, we must have $y_0 \in T$, a contradiction.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.89 第 89 页｜Semi-continuous Correspondences

Semi-continuous Correspondences

We wish to show that $y_0 \in \varphi(x_0)$. Let $y_0 \notin \varphi(x_0)$. Note that $\varphi(x_0)$ is compact, there exist two neighbourhoods $V(y_0)$ and $V'(\varphi(x_0))$ such that $V(y_0) \cap V'(\varphi(x_0)) = \varnothing$, then $y_0 \notin \overline{V'}(\varphi(x_0))$.

If φ is u.s.c. at x_0 and $x_n \to x_0$, then there exists a $N \in \mathbb{N}$ such that $\varphi(x_n) \subseteq V'(\varphi(x_0))$ for $n \geqslant N$, thus $y_n \in \overline{V'}(\varphi(x_0))$ for $n \geqslant N$, and $y_0 \in \overline{V'}(\varphi(x_0))$ for $n \geqslant N$, which contradicts $y_0 \notin \overline{V'}(\varphi(x_0))$.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：在有限维欧氏空间，紧致 $\Leftrightarrow$ 闭且有界 $\Leftrightarrow$ 每个序列有收敛子列且极限仍在集合内。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.90 第 90 页｜Semi-continuous Correspondences

Semi-continuous Correspondences

Proof of (2). Now suppose that for every sequence $x_n \to x_0$ and $y_n \in \varphi(x_n)$, there is a subsequence of y_n with limit in $\varphi(x)$.

Suppose φ is not u.s.c., then there exists an open subset G of Y with $\varphi(x_0) \subseteq G$ and

$$\varphi\left(B_{\frac{1}{n}}(x_0)\right) \setminus G \neq \varnothing$$

for any $n \in \mathbb{N}$. But then there exist $(x_n) \subseteq X$ and $(y_n) \subseteq Y$ such that $x_n \to x_0$ and $y_n \in \varphi\left(B_{\frac{1}{n}}(x_0)\right) \setminus G$ for each n. Since $\varphi(x_0) \subseteq G$ and each y_n belongs to the closed set $Y \setminus G$, no subsequence of (y_n) can converge to a point in $\varphi(x_0)$. □

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.91 第 91 页｜Semi-continuous Correspondences

Semi-continuous Correspondences

Proof of (2). Now suppose that for every sequence $x_n \to x_0$ and $y_n \in \varphi(x_n)$, there is a subsequence of y_n with limit in $\varphi(x)$.

Suppose φ is not u.s.c., then for any open set $G \subseteq Y$ with $\varphi(x_0) \subseteq G$ and any neighbourhood $U_n(x_0)$, there exist $x_n \in U_n(x_0)$ such that $\varphi(x_n) \nsubseteq G$. As $x_n \to x_0$, there exists $y_n \in \varphi(x_n)$ and $y_n \notin G$. Then there exists a convergent subsequence $\{y_{n'}; n' \in \mathbb{N}\} \subseteq \{y_n; n \in \mathbb{N}\}$ such that $y_{n'} \to \varphi(x_0)$. However, $y_n \notin G$, there exists a convergent subsequence $\{y_{n'}; n' \in \mathbb{N}\} \subseteq \{y_n; n \in \mathbb{N}\}$ such that $\lim_{n' \to \infty} y_{n'} \notin G$, then $\lim_{n' \to \infty} y_{n'} \notin \varphi(x_0)$. This is a contradiction! □

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

读课件时的动作： 本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.92 第 92 页｜Properties of Semi-continuous Mappings

Properties of Semi-continuous Mappings

Theorem 3.2. (Inner Semi-continuity) If φ_1 is an i.s.c. mapping of X into Y and φ_2 is an i.s.c. mapping of Y into Z, then the composition product $\varphi = \varphi_2 \cdot \varphi_1$ is an i.s.c. mapping of X into Z.

Proof. If $x_n \to x_0$ and $z_0 \in \varphi(x_0) = (\varphi_2 \cdot \varphi_1)(x_0) = \varphi_2(\varphi_1(x_0))$, then there exists $y_0 \in \varphi_1(x_0)$ such that $z_0 \in \varphi_2(y_0)$.

Whenever $x_n \to x_0$ and $y_0 \in \varphi_1(x_0)$, by inner semi-continuity of φ_1, there exists a sequence $y_n \in \varphi_1(x_n)$ such that $y_n \to y_0$. On the other hand, $z_0 \in \varphi_2 y_0$, by inner semi-continuity of φ_2, there exists a sequence $z_n \in \varphi_2(y_n)$ such that $z_n \to z_0$.

$[y_n \in \varphi_1(x_n)$ and $z_n \in \varphi_2(y_n)]$ imply $z_n \in \varphi_2(\varphi_1(x_n)) = \varphi(x_n)$, thus $\varphi = \varphi_2 \cdot \varphi_1$ is i.s.c. □

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

结合原文理解：读证明先辨认值集是否非空、紧、闭，以及定义域和目标空间的拓扑；缺任何一个假设都可能令结论不成立。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.93 第 93 页｜Properties of Semi-continuous Mappings

Properties of Semi-continuous Mappings

Theorem 3.2′ (Outer Semi-continuity) If φ_1 is an o.s.c. mapping of X into Y and φ_2 is an o.s.c. mapping of Y into Z, the composition product $\varphi = \varphi_2 \cdot \varphi_1$ is an o.s.c. mapping of X into Z.

Proof. If $x_n \to x_0$ and $z_n \in \varphi(x_n) = (\varphi_2 \cdot \varphi_1)(x_n) = \varphi_2(\varphi_1(x_n))$, then there exists $y_n \in \varphi_1(x_n)$ such that $z_n \in \varphi_2(y_n)$.

Whenever $x_n \to x_0$ and $y_n \in \varphi_1(x_n)$, by outer semi-continuity of φ_1, there exists a convergent subsequence $y_{n_1} \to y_0$ with $y_0 \in \varphi_1(x_0)$. On the other hand, $z_n \in \varphi_2(y_n)$, then $z_{n_1} \in \varphi_2(y_{n_1})$. By outer semi-continuity of φ_2, there exists a convergent subsequence $z_{n_2} \to z_0$ with $z_0 \in \varphi_2(y_0)$.

$[y_0 \in \varphi_1(x_0)$ and $z_0 \in \varphi_2(y_0)]$ imply $z_0 \in \varphi_2(\varphi_1(x_0)) = \varphi(x_0)$, thus $\varphi = \varphi_2 \cdot \varphi_1$ is o.s.c. □

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

结合原文理解：读证明先辨认值集是否非空、紧、闭，以及定义域和目标空间的拓扑；缺任何一个假设都可能令结论不成立。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.94 第 94 页｜Semi-continuous Correspondences

Semi-continuous Correspondences

Theorem 3.3 (Inner Semi-continuity) The Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n$ of a finite number of i.s.c. mappings φ_n of X into Y_n is an i.s.c. mapping of X into $Y = \prod_{n=1}^{N} Y_n$.

Proof. If $x^q \to x^0$ and $\left(y_1^0, \cdots, y_N^0\right) = y^0 \in \varphi\left(x^0\right) = \prod_{n=1}^{N} \varphi_n\left(x^0\right)$, then $y_n^0 \in \varphi_n\left(x^0\right)$ for $n = 1, \cdots, N$. If every φ_n is i.s.c., then there is a sequence $y_n^q \in \varphi_n\left(x^q\right)$ for each q such that $y_n^q \to y_n^0$. Thus $y^q = \left(y_1^q, \cdots, y_N^q\right) \in \varphi\left(x^q\right) = \prod_{n=1}^{N} \varphi_n\left(x^q\right)$ such that $y^q = \left(y_1^q, \cdots, y_N^q\right) \to \left(y_1^0, \cdots, y_N^0\right) = y^0$. Therefore $\varphi = \prod_{n=1}^{N} \varphi_n$ is i.s.c. □

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

结合原文理解：读证明先辨认值集是否非空、紧、闭，以及定义域和目标空间的拓扑；缺任何一个假设都可能令结论不成立。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.95 第 95 页｜Semi-continuous Correspondences

Semi-continuous Correspondences

Theorem 3.3′ **(Outer Semi-continuity)** The Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n$ of a finite number of o.s.c. mappings φ_n of X into Y_n is an o.s.c. mapping of X into $Y = \prod_{n=1}^{N} Y_n$.

Proof. If $x^q \to x^0$ and $(y_1^q, \cdots, y_N^q) = y^q \in \varphi(x^q) = \prod_{n=1}^{N} \varphi_n(x^q)$, then $y_n^q \in \varphi_n(x^q)$ for $n = 1, \cdots, N$. If φ_1 is o.s.c., then there is a convergent subsequence $y_1^{q_1} \to y_1^0$ with $y_1^0 \in \varphi_1 x^0$. In this case, $y_n^{q_1} \in \varphi_n x^{q_1}$ for $n = 1, \cdots, N$. If φ_2 is o.s.c., then there is a convergent subsequence $y_n^{q_2} \to y_n^0$ with $y_n^0 \in \varphi_n(x^0)$ for $n = 1, 2$. In this case, $y_n^{q_2} \in \varphi_n x^{q_2}$ for $n = 1, \cdots, N$. Follow these steps, we can obtain that, if φ_N is o.s.c., then there is a convergent subsequence $y_n^{q_N} \to y_n^0$ with $y_n^0 \in \varphi_n(x^0)$ for $n = 1, \cdots, N$. Hence $y^{q_N} = (y_1^{q_N}, \cdots, y_N^{q_N}) \to (y_1^0, \cdots, y_N^0) = y^0$ and $y^0 = (y_1^0, \cdots, y_N^0) \in \prod_{n=1}^{N} \varphi_n(x^0) = \varphi(x^0)$. Therefore $\varphi = \prod_{n=1}^{N} \varphi_n$ is o.s.c. □

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

结合原文理解：读证明先辨认值集是否非空、紧、闭，以及定义域和目标空间的拓扑；缺任何一个假设都可能令结论不成立。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.96 第 96 页｜本页课件

Berge's Maximum Theorem

Theorem 4.1. If f is a l.s.c. numerical function in $S \times T$ and φ is a l.s.c. mapping of S into T such that, for each x, $\varphi(x) \neq \varnothing$, the numerical function M defined by

$$M(x) = \sup\{f(x, y) \mid y \in \varphi(x)\}$$

is l.s.c.

Proof. Suppose that $x_0 \in S$ and let y_0 be such that

$$y_0 \in \varphi(x_0); \qquad f(x_0, y_0) \geqq M(x_0) - \varepsilon.$$

There exist neighbourhoods $U(x_0)$ and $V(y_0)$ such that

$$(x, y) \in U(x_0) \times V(y_0) \Longrightarrow f(x, y) \geqq f(x_0, y_0) - \varepsilon \geqq M(x_0) - 2\varepsilon;$$

there exists a neighbourhood $U'(x_0)$ such that

$$x \in U'(x_0) \Longrightarrow \varphi(x) \cap V(y_0) \neq \varnothing.$$

Therefore

$$x \in U(x_0) \cap U'(x_0) \Longrightarrow M(x) \geqq M(x_0) - 2\varepsilon.$$

□

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 下半连续要求：若 y Φ(x)，当自变量序列趋近给定点时，可以从临近的对应值集中挑选一列点逼近原来的目标点（在适当的度量空间设定下）。

读课件时的动作： 本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

对应的完整证明： 参见本讲义最后的「lsc」证明专题。

2.97 第 97 页｜Berge’s Maximum Theorem

Berge's Maximum Theorem

Theorem 4.2. If f is a u.s.c. numerical function in $S \times T$ and φ is a compact-valued u.s.c. correspondence of S into T such that, for each x, $\varphi(x) \neq \varnothing$, the mumerical function M defined by

$$M(x) = \max\{f(x, y) \mid y \in \varphi(x)\}$$

is u.s.c.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：在有限维欧氏空间，紧致 $\Leftrightarrow$ 闭且有界 $\Leftrightarrow$ 每个序列有收敛子列且极限仍在集合内。

读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

2.98 第 98 页｜本页课件

Berge's Maximum Theorem

Proof. Suppose that $x_0 \in S$; to each y in $\varphi(x_0)$ there correspond neighbourhoods $U_y(x_0)$ and $V(y)$ such that

$$(x,z) \in U_y(x_0) \times V(y) \quad \Longrightarrow \quad f(x,z) \leqq f(x_0,y) + \varepsilon.$$

Since $\varphi(x_0)$ is compact, it can be covered by a finite number of neighbourhoods of the form $V(y)$, say $V(y_1), V(y_2), \cdots, V(y_N)$. Putting $U'(x_0) = \bigcap_{n=1}^{N} U_{y_n}(x_0)$ and $V(\varphi(x_0)) = \bigcup_{n=1}^{N} V(y_n)$, we have

$$x \in U'(x_0), y \in V(\varphi(x_0)) \Longrightarrow f(x,y) \leqq \max_n f(x_0,y_n) + \varepsilon \leqq M(x_0) + \varepsilon.$$

Moreover there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \subseteq V(\varphi(x_0))$$

and so

$$x \in U(x_0) \cap U'(x_0) \quad \Longrightarrow \quad M(x) = \max_{y \in \varphi(x)} f(x,y) \leqq M(x_0) + \varepsilon.$$

□

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 在有限维欧氏空间，紧致 $\Leftrightarrow$ 闭且有界 $\Leftrightarrow$ 每个序列有收敛子列且极限仍在集合内。

读课件时的动作： 本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.99 第 99 页｜Berge’s Maximum Theorem

Berge's Maximum Theorem

Theorem 4.3 (Maximum Theorem). If f is a continuous numerical function in T and φ is a continuous mapping of S into T such that, for each x, $\varphi(x) \neq \varnothing$, then the numerical function M defined by $M(x) = \max\{f(y) \mid y \in \varphi(x)\}$ is continuous in X and the mapping f defined by $f(x) = \{y \in \varphi(x) \mid f(y) = M(x)\}$ is a u.s.c. correspondence of S into T.

Proof. The function f is continuous in $S \times T$ and so M is a continuous numerical function; moreover the mapping Δ given by

$$\Delta x = \{y \in T \mid M(x) - f(y) \leqq 0\}$$

is closed and hence $f = \varphi \cap \Delta$ is u.s.c. $\square$

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：上半连续对应可用“闭集上逆像为闭集”或序列判别，在紧值等假设下成立。不要把闭图像直接当成普遍等价条件。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

对应的完整证明：参见本讲义最后的「usc」证明专题。

2.100 第 100 页｜Berge’s Maximum Theorem

Berge's Maximum Theorem

Let f be a continuous numerical function defined in a topological space T. A family of compact sets $\mathscr{K} = \{K_i \mid i \in I\}$ in T is called selective (with respect to f) if for each i there exists one and only one a_i such that

$$a_i \in K_i; \qquad f(a_i) = \max\{f(y) \mid y \in K_i\}.$$

In other words, the maximum of f is attained at only one point of the set K_i. For example, in $\mathbb{R}^N$ every family of balls is selective, with respect to $f(y) = \pi_i y$; in $\mathbb{R}$, every family of compact sets is selective with respect to $f(y) = y$.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：在有限维欧氏空间，紧致 $\Leftrightarrow$ 闭且有界 $\Leftrightarrow$ 每个序列有收敛子列且极限仍在集合内。

读课件时的动作：本页有例子或构造。自己换一个简单数值例子，算出具体集合或函数值；再思考一旦撤掉某条假设，结论是否还成立。

2.101 第 101 页｜Berge’s Maximum Theorem

Berge's Maximum Theorem

Theorem 4.4. Let φ be a continuous mapping of S into T such that, for each x, $\varphi(x) \neq \varnothing$. If the family of sets $\{\varphi(x) \mid x \in S\}$ is selective, there is a single-valued continuous mapping σ of S into T such that, for each x, $\sigma(x) \in \varphi(x)$.

Proof. Let f be a continuous numerical function in T for which $\{\varphi(x) \mid x \in S\}$ is selective; if $\phi(x) = \{y \in \varphi(x) \mid f(y) = M(x)\}$ then ϕ is a single-valued mapping of S into T. Moreover, it is u.s.c. by the preceding theorem and so is continuous (since it is single-valued). The mapping $\sigma(x) = \phi(x)$ satisfies

$$\sigma(x) \in \varphi(x) \quad \text{for all} \quad x \in S$$

as required. □

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：上半连续对应可用“闭集上逆像为闭集”或序列判别，在紧值等假设下成立。不要把闭图像直接当成普遍等价条件。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

对应的完整证明：参见本讲义最后的「usc」证明专题。

2.102 第 102 页｜Berge’s Maximum Theorem

Berge's Maximum Theorem

Corollary 4.1. If φ is a continuous mapping of X into $\mathbb{R}$ such that, for each x, $\varphi(x) \neq \varnothing$, there exists a continuous single-valued mapping σ such that, for each x, $\sigma(x) \in \varphi(x)$

Proof. It is sufficient to take $f(y) = y$, whence $\sigma(x) = \max \varphi(x)$. □

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：最大值定理的逻辑：非空紧值、连续可行对应 + 连续目标 ⇒ 值函数连续，argmax 对应非空紧值且上半连续。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

对应的完整证明：参见本讲义最后的「berge」证明专题。

2.103 第 103 页｜Berge’s Maximum Theorem

Berge's Maximum Theorem

Theorem 4.5. If f is a l.s.c. numerical function in $S \times T$ and φ is a compact-valued u.s.c. correspondence of S into T such that, for each x, $\varphi(x) \neq \varnothing$, the numerical function M defined by

$$M(x) = \inf\{f(x, y) \mid y \in \varphi(x)\}$$

is l.s.c.

Proof. We want to prove that, for any $\varepsilon > 0$, there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \Longrightarrow M(x_0) - \varepsilon \leqslant M(x).$$

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：在有限维欧氏空间，紧致 $\Leftrightarrow$ 闭且有界 $\Leftrightarrow$ 每个序列有收敛子列且极限仍在集合内。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.104 第 104 页｜ Berge’s Maximum TheoremIf φ is compact-valued, then the

Berge's Maximum Theorem

If φ is compact-valued, then there exist $y_1, \cdots, y_K \in \varphi(x_0)$ such that $\varphi(x_0) \subseteq \bigcup_{k=1}^{K} V_k(y_k)$. If φ is u.s.c., then there exists a neighbourhood $U_0(x_0)$ such that

$$x \in U_0(x_0) \Longrightarrow \varphi(x) \subseteq \bigcup_{k=1}^{K} V_k(y_k).$$

If f is l.s.c., then, for any $\varepsilon > 0$, $y_k \in \varphi(x_0)$, there exists a neighbourhood $U_{y_k}(x_0)$ such that, for any $x \in U_{y_k}(x_0)$ and any $y \in V_k(y_k)$, $f(x_0, y_k) - \varepsilon \leqslant f(x, y)$.

Define $U(x_0) = U_0(x_0) \cap \bigcap_{k=1}^{K} U_{y_k}(x_0)$. Note that $y_1, \cdots, y_K \in \varphi(x_0)$, then, for any $x \in U(x_0)$ and $y \in \varphi(x)$,
$M(x_0) - \varepsilon \leqslant \inf_{y_k \in \varphi(x_0)} f(x_0, y_k) - \varepsilon \leqslant f(x, y)$.

Therefore, for any $x \in U(x_0)$ and $y \in \varphi(x)$,
$M(x_0) - \varepsilon \leqslant \inf_{y \in \varphi(x)} f(x, y) = M(x_0)$. □

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 在有限维欧氏空间，紧致 ⇔ 闭且有界 ⇔ 每个序列有收敛子列且极限仍在集合内。

读课件时的动作： 本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.105 第 105 页｜Berge’s Maximum Theorem

Berge's Maximum Theorem

Theorem 4.6. If f is a u.s.c. numerical function in $S \times T$ and φ is a l.s.c. mapping of S into T such that, for each x, $\varphi(x) \neq \varnothing$, the numerical function M defined by

$$M(x) = \inf\{f(x, y) \mid y \in \varphi(x)\}$$

is u.s.c.

Proof. We want to prove that, for any $\varepsilon > 0$, there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \Longrightarrow M(x) \leqslant M(x_0) + \varepsilon.$$

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：上半连续对应可用“闭集上逆像为闭集”或序列判别，在紧值等假设下成立。不要把闭图像直接当成普遍等价条件。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

对应的完整证明：参见本讲义最后的「usc」证明专题。

2.106 第 106 页｜Berge’s Maximum Theorem

Berge's Maximum Theorem

From the definition of infimum, for any $\varepsilon > 0$, there exists $y_0 \in \varphi(x_0)$ such that $f(x_0, y_0) \leqslant M(x_0) + \frac{\varepsilon}{2}$.
If f is u.s.c., then there exist two neighbourhoods $U_1(x_0)$ and $V(y_0)$ such that, for any $x \in U_1(x_0)$ and any $y \in V(y_0)$, $f(x, y) \leqslant f(x_0, y_0) + \frac{\varepsilon}{2}$.
Therefore $f(x, y) \leqslant f(x_0, y_0) + \frac{\varepsilon}{2} \leqslant M(x_0) + \varepsilon$.
If φ is l.s.c., for $\varphi(x_0) \cap V(y_0) \neq \varnothing$, then there exists a neighbourhood $U_2(x_0)$ such that

$$x \in U_2(x_0) \Longrightarrow \varphi(x) \subseteq \varphi(x) \cap V(y_0) \neq \varnothing.$$

Define $U(x_0) = U_1(x_0) \cap U_2(x_0)$. for any $x \in U(x_0)$, there exists $y \in V(y_0)$, such that

$$M(x) = \inf_{y \in \varphi(x)} f(x, y) \leqslant f(x, y) \leqslant f(x_0, y_0) + \frac{\varepsilon}{2} \leqslant M(x_0) + \varepsilon.$$

Therefore M is u.s.c. □

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：下确界是所有下界中最大的一个。最好先画数轴，再将量词写成 a A 与 ε>0。

读课件时的动作：本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.107 第 107 页｜Berge’s Maximum Theorem

Berge's Maximum Theorem

Theorem 4.7 (Maximum Theorem). If f is a continuous numerical function in T and φ is a continuous mapping of S into T such that, for each x, $\varphi(x) \neq \varnothing$, then the numerical function M defined by $M(x) = \min\{f(y) \mid y \in \varphi(x)\}$ is continuous in X and the mapping f defined by $\phi(x) = \{y \in \varphi(x) \mid f(y) = M(x)\}$ is a u.s.c. correspondence of S into T.

Proof. The function f is continuous in $S \times T$ and so M is a continuous numerical function; moreover the mapping Δ given by

$$\Delta x = \{y \in T \mid M(x) - f(y) \leqq 0\}$$

is closed and hence $\phi = \varphi \cap \Delta$ is u.s.c. $\square$

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：上半连续对应可用“闭集上逆像为闭集”或序列判别，在紧值等假设下成立。不要把闭图像直接当成普遍等价条件。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

对应的完整证明：参见本讲义最后的「usc」证明专题。

2.108 第 108 页｜Berge’s Maximum Theorem

Berge’s Maximum Theorem

In economic activities, decision makers often change their preferences according to the situation of the environment, so the preference in economic activities is often changed by the environment on which preferences depend. Suppose $\beta : S \to T$ is a set-valued mapping, for any $x \in S$, $\prec_x$ is a binary relationship defined on $\beta(x)$ (set-valued mapping), which is transitive, but is not reflexive and complete. Set $\varphi(x)$ to be the set of maximal elements in $\beta(x)$ according to $\prec_x$.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：最大值定理的逻辑：非空紧值、连续可行对应 + 连续目标 ⇒ 值函数连续，argmax 对应非空紧值且上半连续。

读课件时的动作：本页有例子或构造。自己换一个简单数值例子，算出具体集合或函数值；再思考一旦撤掉某条假设，结论是否还成立。

对应的完整证明：参见本讲义最后的「berge」证明专题。

2.109 第 109 页｜Berge’s Maximum Theorem

Berge's Maximum Theorem

Theorem 4.8. Suppose that $\beta : S \to T$ is a correspondence, $\varphi(x) = \left\{ y \in T \mid y \in \max_{\prec_x} \beta(x) \right\}$, and $\{(x, y, z) \in S \times T \times T \mid y \in \beta(x), z \in \beta(x), y \not\prec_x z\}$ is closed. If $\beta(x)$ is compact, then $\varphi(x) \neq \varnothing$ is compact, β is continuous, φ is u.s.c.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：在有限维欧氏空间，紧致 $\Leftrightarrow$ 闭且有界 $\Leftrightarrow$ 每个序列有收敛子列且极限仍在集合内。

读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

2.110 第 110 页｜Berge’s Maximum Theorem

Berge's Maximum Theorem

Proof. If $\varphi(x) = \varnothing$, then for any $y \in \beta(x)$, there exists $y' \in \beta(x)$ such that $y \prec_x y'$. Define $r(z) = \{y \in \beta(x) \mid y \prec_x z\}$, then $\beta(x) \subseteq \bigcup_{z \in \beta(z)} \{r(z)\}$. Note that $\beta(x)$ is compact and $\bigcup_{z \in \beta(z)} \{r(z)\}$ is its open covering, then there exist $r(z_1), \cdots, r(z_K)$, such that $\beta(x) \subseteq \bigcup_{k=1}^{K} r(z_k)$. $\{z_1, \cdots, z_K\}$ in finite, then there is the maximal to be z_K, as $\prec_x$ is transitive, then for $k = 1, \cdots, K-1$, $z_K \notin r(z_k)$. But $\prec_x$ is not reflexive, then $z_K \notin r(z_K)$, and hence $z_K \notin \beta(x)$, which is a contradiction. Therefore $\varphi(x) \neq \varnothing$.

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 在有限维欧氏空间，紧致 $\Leftrightarrow$ 闭且有界 $\Leftrightarrow$ 每个序列有收敛子列且极限仍在集合内。

读课件时的动作： 本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.111 第 111 页｜Berge’s Maximum TheoremAccording to the continuity of

Berge's Maximum Theorem

According to the continuity of $\prec_x$, $\{y \in \beta(x) \mid y \not\prec_x z\}$ is closed, then

$$\varphi(x) = \{y \in T \mid z \in \beta(x), y \not\prec_x z\} = \bigcap_{z \in \beta(x)} \{y \in \beta(x) \mid y \not\prec_x z\}$$

is closed, $\varphi(x) \neq \varnothing$ and $\varphi(x) \subseteq \beta(x)$, $\beta(x)$ is compact, then $\varphi(x)$ is compact.

In order to check that φ is u.s.c.at x_0, we want to prove that φ is closed-valued,

$$\text{for any } (x_n, y_n) \to (x, y),\ y_n \in \varphi(x_n) \quad \Longrightarrow \quad y \in \varphi(x).$$

Since β is l.s.c., then, for $z \in \beta(x)$, there exists $z_n \to z$ such that, for any n, $z_n \in \varphi(x_n)$, then $y_n \not\prec_{x_n} z_n$. From the continuity of $\prec_x$, $x_n \to x$, $y_n \to y$, $z_n \to z$, then $y \not\prec_x z$, and hence $y = \max_{\prec_x} \beta(x)$, that is, $y \in \varphi(x)$, thus φ is u.s.c. $\square$.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：在有限维欧氏空间，紧致 $\Leftrightarrow$ 闭且有界 $\Leftrightarrow$ 每个序列有收敛子列且极限仍在集合内。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.112 第 112 页｜Berge’s Maximum Theorem

Berge's Maximum Theorem

Corollary 4.2. If f is a continuous numerical function in $S \times T$ and φ is a continuous mapping of S into T such that, for each x, $\varphi(x) \neq \varnothing$, the numerical function M defined by

$$M(x) = \max\{f(x, y) \mid y \in \varphi(x)\}$$

is continuous in X and the mapping ϕ defined by

$$\phi(x) = \{y \in \varphi(x) \mid f(y) = M(x)\}$$

is u.s.c. of S into T.

这一页怎么读

概念主线：对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点：上半连续对应可用“闭集上逆像为闭集”或序列判别，在紧值等假设下成立。不要把闭图像直接当成普遍等价条件。

读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

对应的完整证明：参见本讲义最后的「usc」证明专题。

2.113 第 113 页｜ Berge’s Maximum Theorem

Berge's Maximum Theorem

Proof. Define

$$y \prec_x z \qquad \Longleftrightarrow \qquad f(x,y) \prec f(x,z)$$

Since f is continuous, then $\{(x,y,z) \in S \times T \times T \mid f(x,y) \not\prec f(x,z)\}$ is closed, then for any $x \in S$ and $y \in Y$, $\{z \in Y \mid f(x,y) \prec_x f(x,z)\}$ is closed, then f is u.s.c.

We now check that M is continuous. $\{x\}$ is a single-point mapping of correspondence, the product mapping $(x, r(x))$ is u.s.c., f is continuous, then $f(x, r(x))$ is u.s.c., but $f(x, r(x)) = \{M(x)\}$, then $\alpha : S \to \mathbb{R}$ is continuous. $\square$

这一页怎么读

概念主线： 对应的连续性分为上半连续与下半连续。前者直观地禁止“突然冒出远处的可行点”，后者禁止“旧点突然消失”。

本页重点： 上半连续对应可用“闭集上逆像为闭集”或序列判别，在紧值等假设下成立。不要把闭图像直接当成普遍等价条件。

读课件时的动作： 本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

对应的完整证明： 参见本讲义最后的「usc」证明专题。

3 证明专题：考试应会的标准推导

专题的使用方法

这些专题沿着本课 PPT 的定理顺序补足主要推导；若原课件的命题另有附加假设，以该页原文为主，切勿将证明中的条件删掉。对于高维拓扑不动点定理，只陈述严格的适用条件并说明所需工具，不伪称给出了没有依据的完整高维证明。

3.1 专题：usc

证明｜上半连续对应的序列逻辑（紧值版本）

设 X, Y 为度量空间，Y 紧，$\Phi : X \rightrightarrows Y$ 非空闭值。假定 Φ 上半连续。若 $x_n \to x$，$y_n \in \Phi(x_n)$，由 Y 紧可抽出 $y_{n_k} \to y$。若 $y \notin \Phi(x)$，利用 $\Phi(x)$ 闭及上半连续性，可以找围绕 $\Phi(x)$、不含 y 的开邻域，并得充分接近 x 时 $\Phi(x_n)$ 也落在该邻域内，这与 $y_{n_k} \to y$ 矛盾，所以 $y \in \Phi(x)$。这里目标空间紧使抽子列的步骤成立；没有适当紧性时不能随便用此论证。

3.2 专题：lsc

证明｜下半连续对应的直观反例

定义 $\Phi(x) = \{0\}$（$x \neq 0$），$\Phi(0) = \{0, 1\}$。在 $x = 0$ 处，取 $y = 1 \in \Phi(0)$ 和 $x_n = 1/n \to 0$，但 $\Phi(x_n) = \{0\}$，不可能选 $y_n \in \Phi(x_n)$ 使 $y_n \to 1$，所以下半连续性失败。另一方面，这一对应在 0 处上半连续：其附近的值都在 $\{0, 1\}$ 的任意开邻域中。这说明上半连续与下半连续不等价。

3.3 专题：berge

证明｜Berge 最大值定理：值函数连续的证明结构

设 X, Y 为度量空间，$\Gamma : X \rightrightarrows Y$ 非空紧值且连续，$u : X \times Y \to \mathbb{R}$ 连续，$v(x) = \max_{y \in \Gamma(x)} u(x, y)$。取 $x_n \to x$。对每个 n 取最优 $y_n \in \Gamma(x_n)$。由局部紧性/紧值上半连续提供的序列紧性（本课程的有限维、局部紧条件下），取子列 $y_{n_k} \to y \in \Gamma(x)$，得 $\limsup v(x_n) \leq u(x, y) \leq v(x)$。另一方面取 $y^* \in \arg\max_{\Gamma(x)} u(x, \cdot)$，下半连续性允许选 $z_n \in \Gamma(x_n)$ 使 $z_n \to y^*$，故 $\liminf v(x_n) \geq \lim u(x_n, z_n) = v(x)$。两侧合并得到 $v(x_n) \to v(x)$。极大点对应的上半连续性再由最优性不等式和闭图像/紧性推出。

3.4 专题：evt

证明｜连续函数在非空紧集上取得最大、最小值

设 $K \subset \mathbb{R}^N$ 非空紧致，$f : K \to \mathbb{R}$ 连续。先证有界：若无上界，取 $x_n \in K$ 使 $f(x_n) > n$。紧致性给出收敛子列 $x_{n_k} \to x \in K$，连续性给出 $f(x_{n_k}) \to f(x)$，与 $f(x_{n_k}) > n_k \to \infty$ 矛盾。令 $M = \sup_{x \in K} f(x)$，依上确界性质取 $x_n \in K$ 使 $f(x_n) > M - 1/n$，取收敛子列 $x_{n_k} \to x^* \in K$，连续性推出 $f(x^*) = M$。对 $-f$ 重复论证即得最小值。

4 自测清单

1. 你能不看课件写出本讲的核心定义及所有量词吗?
2. 你能逐条列出每一个定理的假设，并解释删去一条为何可能失败吗?
3. 你能在白纸上独立复述本讲证明专题中的每一个箭头吗?
4. 你能结合原课件的反例判断是否把充分条件错写成必要条件吗?